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REVIEW

Advancements and Theoretical Foundations of Cable-Driven Parallel Robots: A Comprehensive Overview

Kamran Joyo[1,2,4], Han Yuan[1]*, Jun Wu[3]*, Wenfu Xu[1], Honghao Yue[3]

Abstract

Parallel robotic mechanisms using cables instead of rigid limbs are termed cable-driven parallel robots (CDPRs). Electric motors and pulley mechanisms actuate cables to provide motion for an end-effector in a cable robot. Consequently, CDPRs have emerged as indispensable tools across a spectrum of industrial and technological domains, including astronomy, aerospace, logistics, simulators, and rehabilitation. Their inherent compatibility with the evolving concept of rigid-flexible fusion places CDPRs at the forefront of cutting-edge robotics research. This comprehensive paper aims to consolidate the core theories and advancements underpinning CDPRs, encompassing key aspects such as configuration design, cable-force distribution, workspace and stiffness analysis, performance evaluation, optimisation techniques, and motion control. We provide in-depth insights into kinematic modelling, workspace exploration, and cable-force solutions. Furthermore, the paper delves into the intricacies of stiffness and dynamic modelling, presenting a range of analytical methods to elucidate their effects on CDPR performance. Addressing reliability concerns and developing a unified control framework are identified as essential in ensuring the practical deployment of CDPRs in real-world scenarios. This research paper offers a comprehensive overview of the theories and advancements in CDPRs, identifying critical areas for further research and development to unlock the full potential of these versatile and high-performance robotic systems.

Keywords: Cable-driven parallel robots, Optimisation, Path generation, Statics, Machine Learning

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1 Introduction

A parallel robot with a cable-driven motor is called a cable-driven parallel robot (CDPR). Traditionally parallel robots are equipped with rigid links, whereas the moving platform of CDPR is suspended by several cables. In contrast to conventional link-based robots, CDPRs offer improved accelerations and velocities due to their lower inertia, spacious workspaces, and exceptional payload capacities [1]. The earliest ideas about robotic systems that are controlled by cables originated in the 1980s. Developed in the 1980s for performing pick-and-place and carrying a recording camera, respectively, SkyCam and RoboCrane are considered the first robots of this kind [2,3]. In the 1990s, researchers pioneered the concept of augmenting Cable-Driven Parallel Robots (CDPRs) with additional cables as a novel approach to address tension distribution challenges [4]. Subsequently, the classification of CDPRs based on cable count and degrees of freedom (DOF) was initially introduced in a seminal work [5]. Over the years, CDPRs have gained increasing prominence in both academia and industry, with extensive research contributing to their widespread adoption in diverse applications, such as pick-and-place operations [6], therapeutic devices [7], and advanced 3D printing technology [8]. Table 1 provides summaries of literature outlines sourced from recent critical review sources. These summaries highlight key insights and findings from a variety of scholarly sources, offering valuable perspectives on the subject of CDPRs.

There are a number of variables that can affect the static and dynamic performance of a CDPR, including the materials, density, and tensile strength of the cables [9]. CDPR reliability is often examined in studies for accuracy based on the mass and elasticity of cables. Whenever the tension of a cable is substantially greater than its weight, it is possible to ignore its mass [10]. Nevertheless, in applications with large workspaces, a large cable mass needs to be accounted for when formulating motion

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dynamics [11]. For instance, Chinese scientists designed the world's largest single-dish radio telescope, the five-hundred meter aperture spherical radio telescope (FAST). A 6-cable CDPR mechanism with a 600-meter span supports the FAST feed system illustrated in Figure 1(a) [12,13]. It is also worth mentioning MARIONET-CRANE illustrated in Figure 1(b) [14,15], COGIRO illustrated in Figure 1(c) [16] and Façade greening illustrated in Figure 1(d) [17] designed to handle and assemble bulky parts. With the applications illustrated in Figure 1, cable modelling with mass should be added to the consideration of kinematics and dynamics equations for effective analysis. CDPR performance and efficiency are greatly influenced by cable elasticity, especially when dealing with high-speed applications [18,19].

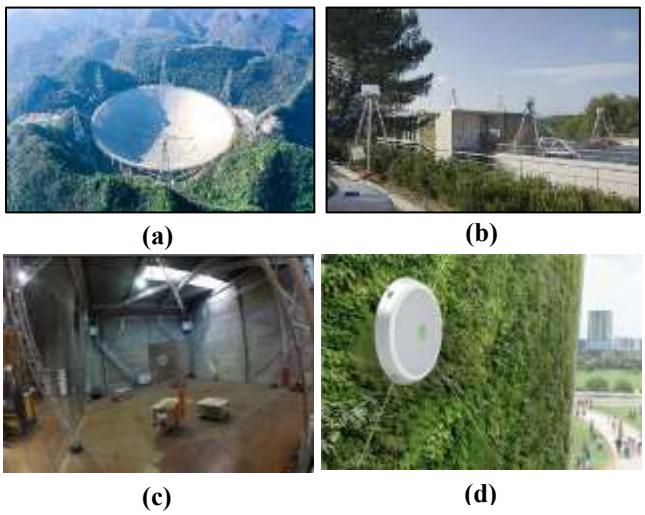

Figure 1 Structures of CDPRs: (a) FAST, (b) MARIONET-CRANE, (c) COGIRO, (d) Façade greening

A CDPR is capable of generating accelerations exceeding 400 m/s^2 [9]. In Ref. [20], a high-speed spatial CDPR with the ability to accelerate up to 43 m/s^2 and travel at a speed of 13 m/s was introduced. Recent experiments have shown that CDPR can accelerate to approximately 31.81 m/s at a maximum speed of 66.2778 m/s^2 [19]. When formulating CDPR's kinematics and dynamics, cables' elasticity should be considered to determine its effects on displacements and vibrations produced.

In contrast to rigid robots, CDPRs are more reconfigurable and versatile due to their lightweight structure and modularity. Extensive research in the existing literature has investigated into a diverse array of cable and motor configurations with the aim of achieving optimal path planning [21–24] as well as maximising workspace generation [21,25–27]. A cluttered environment can also result in cables wrapping around obstacles or colliding with each other. Considering the possibility of preventing cables from colliding, Refs. [21] and [25] have contemplated modifying the configuration of their platform by changing the connection points.

Table 1 A summary of the literature from various recent critical review sources

Ref.	Purpose	Type of Ref.	Summary points
Qian et al. [1]	Review CDPR development, theory, and applications.	Review	Review introduces CDPR history and recent advances, discussing design, performance analysis, and control theory. Highlights CDPR significance in modern engineering.
Pott [9]	Comprehensive theory of CDPRs.	Book	Provides an extensive overview of CDPRs, covering kinematic and dynamic analysis, design features, and operational capabilities. Explores nonstandard cable models and diverse applications.
Tho et al. [28]	Define workspace, distribution of cable tensions and cable sagging.	Review	The study serves as a valuable resource by summarizing published studies and categorizing computational methods used for CDPR design and control. The efficacy of these computational methods is assessed based on mathematical theories governing CDPR kinematics and dynamics. The findings of this study provide scholars with a comprehensive reference to expedite their investigations into critical problems related to CDPR design and implementation.
Zhang et al. [29]	Overview of CDPR theories and applications.	Review	Thie study discusses kernel theories and developments of CDPRs with an emphasis on configuration design, cable force distribution, workspace, and stiffness. It also discusses performance evaluation with workspace optimization, and motion control.
Zarebidoki et al. [30]	Review CDPR configurations, analysis, and control methods.	Review	Summarizes CDPR configurations, emphasizing the impact of cable mass and elasticity. Discusses workspace optimization, trajectory planning, and control algorithms.

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Present Study	Comprehensive qualitative and quantitative overview of development, applications, and theoretical foundations of CDPR.	Review	Analyzes design configurations, modeling techniques, and control systems for CDPRs. Emphasizes recent methodologies for underactuated CDPRs and highlights learning-based intelligent controls for enhanced efficiency and adaptability.

This review article on CDPRs provides a comprehensive qualitative and quantitative overview of their development, applications, and theoretical foundations. It highlights the advantages of CDPRs over rigid-link robots, such as larger workspaces and higher payload-to-weight ratios. The paper discusses the history of CDPR development and presents recent successful application cases. Notably, the paper delves into recent advancements in underactuated CDPRs, elucidating their theoretical progress and practical implications. It emphasizes the need for further research in areas like intelligent control, composite materials, and reconfigurable design to meet evolving engineering demands. The review concludes by outlining future research directions, including integrated design, advanced intelligent controllers, and modular design, to enhance CDPR performance and applicability across various fields. This study's unique contribution lies in its in-depth coverage of recent studies on underactuated CDPRs, and integration learning algorithms, enriching the discourse on their theoretical advancements and practical applications within the CDPRs landscape.

Initially, this research summarizes some common CDPR configurations and the applications they are used for. Further, discuss various methods for improving workspace geometry, such as integrating actuators and cables into CDPRs or optimising cable arrangements. Following that, several cable models with mass and elasticity are presented in order to integrate them into the equations of motion. Additionally, differentiated approaches to path planning for both planar and spatial CDPRs are reviewed, along with methods for enhancing design configurations in order to achieve collision-free path planning. Further, control algorithms are presented that consider the cable mass and flexibility. The literature includes methods to cope with structured and unstructured uncertainties, such as the mass and moment of end-effector and external disturbances.

2 Background

Researchers are mainly interested in CDPRs related to the main contents, which include structural optimization, determining and enhancing working space, strategies for distributing cable tension, assessing cable sagging models, and developing controllers for motion trajectory.

It is most common to classify CDPRs according to their structure, the number of DOF, and the number of cables, with 'n' being the number of DOF and 'm' being the number of cables used to drive the device [28]. Recent surveys have pulled all of these strands together, showing how CDPR research has matured across configuration design, workspace evaluation, stiffness modeling, and control challenges [29,30]. CDPRs are typically configured as follows [28].

A CDPR is under constrained or incompletely restrained positioning mechanism (IRPM), if $m \leq n$. If the robot is configured in such a way, it cannot withstand wrenches applied arbitrarily. The robot may, however, be in stable or unstable equilibrium in one or more poses of an IRPM when gravity is considered, or other forces are applied.

For $m = n + 1$, certain positions allow the CDPR to act as a completely restrained positioning mechanism (CRPM). For specific poses, addressing tension distribution becomes relatively straightforward, given that it establishes a one-dimensional subspace for cable forces.

For $m > n + 1$, the CDPR is said to be in a redundantly restrained positioning mechanism (RRPM). There might be an infinite number of tension distribution solutions based on the robot configuration in this case; therefore, two significant problems need to be addressed. For a tension distribution the initial step is to attempt to identify single solution. In the event that there is more than one solution, the second problem is to identify continuous solutions for the given path [31]. In Figure 2, CDPR arrangements are illustrated [9]. Verhoeven[32] developed a method for CDPR classification, which took into account cable constraint capacity. This established a relationship between the number of driving cables and the number of terminal DOF.

Besides the standard structure of CDPRs, various configurations have also been suggested that replace some cables with rigid links with retractable mechanisms, or springs and hinge-type joints. In Ref. [33], all cables are maintained in tension by telescopic elements that retract onto the mobile platform. The addition of an actuator will increase the force and torque performance of the mobile platform when the required ballast force is applied. Mechanisms driven by cables with planar motion were presented in Ref. [34], where the actuators of the robot are controlled by a cable loop connected to a rigid link cartesian mechanism. The structure of CDPRs has also been designed

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with springs in order to constrain the robot while eliminating the requirement for additional cables and driving components [35] or to improve and optimise the workspace [36,37]. We examine the kinematics, dynamics, workspaces, path planning, and control aspects of CDPRs in different configurations and arrangements in the following sections.

Several typical applications corresponding to different configurations of CDPRs are illustrated in Figure 3 which are also enlisted in Table 2. Figure 3 has been separated into two parts in order to simplify the figure where axes represent the year of development and cable configuration. The cable configuration has been represented numerically (1. Non-Flexible Cables without mass, 2. non-flexible cables with mass, 3. Massless flexible cables, 4. Hefty and Flexible cables). The size of the bubble also presented numerically represents the CDPR configuration (1. planar RRPM, 2. planar CRPM, 3. planar IRPM, 4. spatial RRPM, 5. spatial CRPM, 6. spatial IRPM). The data points indicate a progression in the use of different CDPR configurations over the years, with a noticeable shift towards more complex configurations (e.g., spatial RRPM) as the technology matures. This suggests an evolving interest in exploring more sophisticated and versatile CDPRs in various applications.

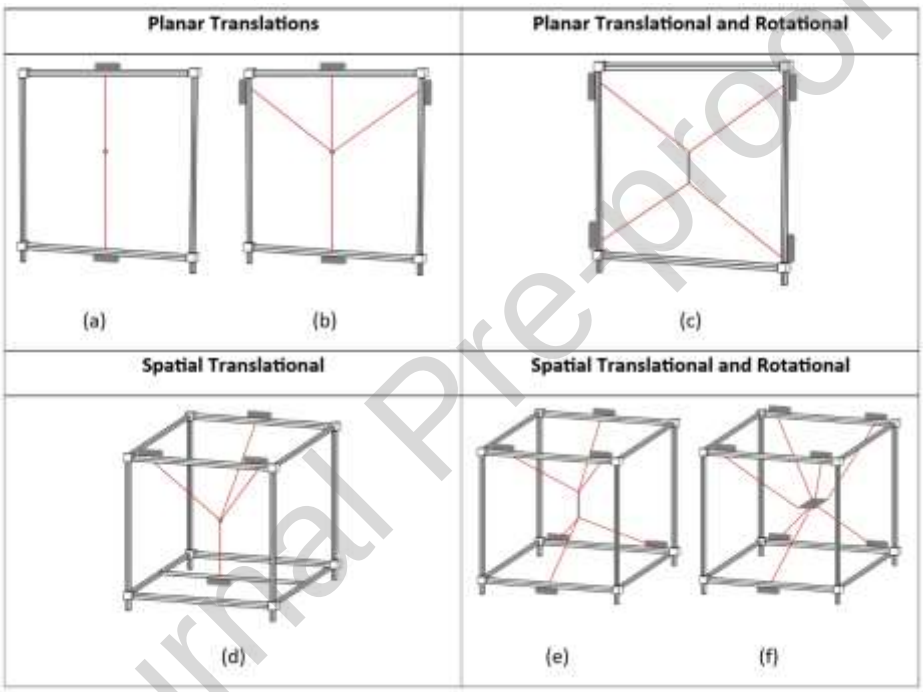

Figure 2 Possible arrangements of CDPRs: (a) Arrangement 1: 2-1 planar CRPM, (b) Arrangement 2: 4-2 planar RRPM, (c) Arrangement 3: 4-3 planar CRPM, (d) Arrangement 4: 4-3 spatial CRPM, (e) Arrangement 5: 6-5 spatial CRPM, (f) Arrangement 6: 7-6 spatial CRPM

Table 2 **CDPR applications corresponding to its arrangement**

Applications	Arrangement	References
Pick and place	Planar RRPM	[6,19,38]
	Planar CRPM	[39]
	Spatial IRPM	[25,40–47]
	Spatial CRPM	[48]
	Spatial RRPM	[49,50]
Large aperture telescope	Spatial IRPM	[12,51,52]
Rehabilitation	Planar CRPM	[53–66]
	Planar IRPM	[67]
	Spatial IRPM	[68–71]
	Spatial CRPM	[72–76]
Printing (3D and structural)	Planar RRPM	[6,77]
	Spatial IRPM	[78,79]
	Spatial CRPM	[80–83]

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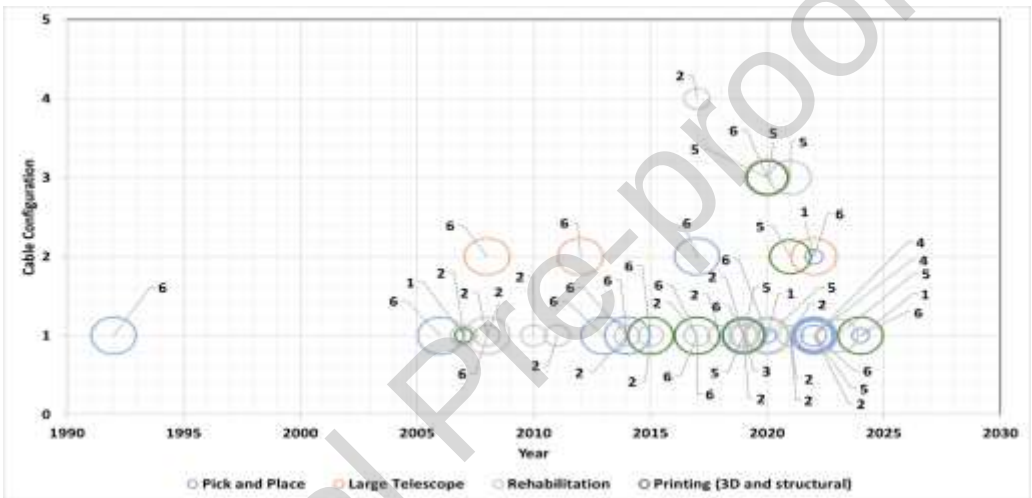
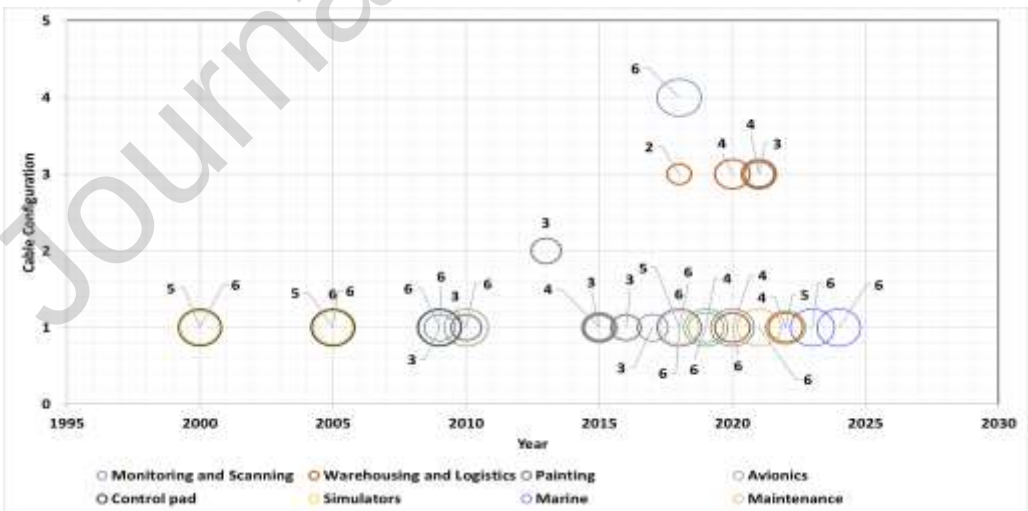
Monitoring and scanning	Planar IRPM	[84,85]
	Spatial IRPM	[86–89]
Warehousing and logistics	Planar CRPM	[90,91]
	Spatial RRPM	[50,92–94]
Painting	Planar IRPM	[53,95–98]
	Spatial RRPM	[21]
Avionics	Spatial IRPM	[99–102]
	Spatial CRPM	[48]
	Spatial RRPM	[103]
Control pad	Spatial IRPM	[104–106]
Maintenance	Spatial IRPM	[17]
Marine	Spatial IRPM	[107,108]
	Spatial RRPM	[109]
Simulators	Spatial CRPM	[104,105,108,110]



(a)



(b)

Figure 3 Studies reporting applications corresponding to configuration of CDPRs. Colour represents the type of application; size represents the CDPR's configuration (1. planar RRPM, 2. planar CRPM, 3. planar IRPM, 4. spatial RRPM, 5. spatial CRPM, 6. spatial IRPM); Axes represents their year of development and cable configuration (1. Non-Flexible Cables without mass, 2. non-flexible cables with mass, 3. Massless flexible cables, 4. Hefty and Flexible cables). (a) Cable configurations for applications: Pick and place, large telescope, rehabilitation and 3D printing, (b) Cable configuration for applications: Monitoring, logistics, painting, avionics, control pad, simulation, marine and maintenance

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3 Materials and Methods

3.1 Kinematics and Dynamics

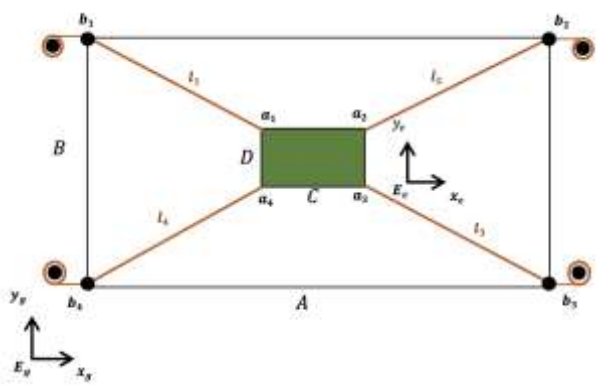

Figure 4 Typical kinematic diagram of the CDPR

The forward kinematics (FK) of a parallel robot presents a significantly more intricate challenge for solution compared to that of a serial robot. This complexity is further amplified when dealing with cable-driven parallel robots (CDPRs) due to their distinctive design requirements, particularly the necessity of maintaining cable tension at all times to ensure operational stability and precision. In the case of inverse kinematics (IK) involving parallel-configured robots and CDPRs with cables that are either massless or inelastic, the computational burden is notably reduced. This reduction stems from the simplified nature of these cables, which allows for relatively straightforward solutions to determine the end-effector's position and orientation based on the desired task. However, an additional layer of complexity emerges when grappling with the inverse kinematic problem for CDPRs that employ hefty cables with elastic properties. In this scenario, several factors contribute to the heightened intricacy of the problem. Firstly, the exact position variables and cable tensions are not known beforehand, necessitating the simultaneous determination of these parameters. Secondly, the mathematical expressions governing the behaviour of these cables are non-algebraic, making the problem challenging to solve analytically. The challenge is further intensified with the hefty and elastic cables. The tensioning cables in these configurations exhibit a dual nature, influenced by both their substantial physical properties and their elasticity. As a result, devising a comprehensive solution to the inverse kinematic problem for CDPRs with such cables becomes a progressively intricate task. The need to account for both the bulkiness and elasticity of the cables in the kinematic calculations underscores the need for advanced computational methods and numerical techniques to arrive at accurate and reliable solutions [111].

3.1.1 Non-Flexible Cables without Mass

This section delves into an in-depth exploration of nonelastic cable models and cables with zero mass, illuminating their profound implications for the kinematics and dynamics of CDPRs. Understanding the behaviour of such cables is pivotal to developing accurate models and predictive frameworks for these sophisticated robotic systems. Nonelastic cable models, which disregard the inherent elasticity of cables, play a crucial role in CDPR analysis. By considering cables as perfectly rigid elements, these models simplify the mathematical representation of the robot's behaviour, offering a basis for precise kinematic and dynamic calculations. It is quite common to use a straight-line model for kinematics modelling. Referring to Figure 4, the length of each cable can be determined from the pose of end-effector.

$$\ell_i = \|b_{gi} - a_{gi}\|, \tag{1}$$

where ℓ_i represents the length of ith cable, a_{gi} is the position of cable attachment at the end-effector while b_{gi} is the attachment point at the base. This simplification is particularly valuable when rapid and deterministic solutions are sought, allowing for the efficient determination of cable tensions and end-effector positions. Despite being conceptually useful in modelling CDPRs, the straight-line model tends to overlook two important practical considerations: Pulleys within cable assemblies and the cable's inherent shape. The importance of these factors lies in the accurate representation of CDPR behaviour and in tackling real-world challenges associated with them. As a method for solving dynamics equations, Newton-Euler has proven to be helpful for both spatial and planar CDPRs [41,112–114]. This method provides a robust framework for modelling the dynamic behaviour of CDPRs, enabling the calculation of position vectors for each cable and the determination of applied forces and wrenches on the end-effector. By utilizing the Newton-Euler method, it becomes possible to derive comprehensive and accurate dynamics equations, shedding light on the intricate interplay of forces and motion within CDPR. As the basis of the analysis framework, the Lagrange method serves as a pivotal tool for constructing Jacobian matrices. This Jacobian matrix is systematically assembled by leveraging a set of variables derived from an in-depth dynamics analysis, as outlined in Ref. [115]. This approach offers a systematic and rigorous

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means of characterizing the intricate relationships between robot kinematic variables and its dynamic properties. In the context of this framework, Ref. [113], delineates the identification of six parameters, each corresponding to one of the six degrees of freedom inherent to the end-effector, encompassing both positional and orientational aspects. These parameters play a fundamental role in defining the robot's capabilities, enabling it to navigate within its operational space. Moreover, the Jacobian matrix, an integral part of the updated workspace definition, capitalizes on the Cartesian axes of the end-effector. The approach is particularly notable for reducing computational complexity and enhancing efficiency. By incorporating nine variables into the matrix derived from the end-effector's vertices, computational time is significantly streamlined. Direct kinematic calculations based solely on cable-length geometric constraints were presented for a planar 2-3 IRPM. Three sensor fusion algorithms (norm, strict and engineering Gauss-Newton) are introduced, based on nonlinear weighted least squares equations, each differing in termination conditions. Experimentally, various redundant measurement sets are compared for optimal pose estimation. Results indicate that measurements of cable orientation degrade accuracy in determining platform poses, while measures of cable lengths and platform orientation significantly enhance accuracy. The proposed termination condition for the Gauss-Newton method reduces computation time while maintaining accuracy and offers a meaningful physical interpretation. Moreover, different IRPM configurations may require varying sensor types based on their degree of underactuation. A multilayer perception neural network approach was used to study the forward kinematics in [116], which is less computationally intensive than a numerical technique. Training the network was accomplished through the back-propagation method. It

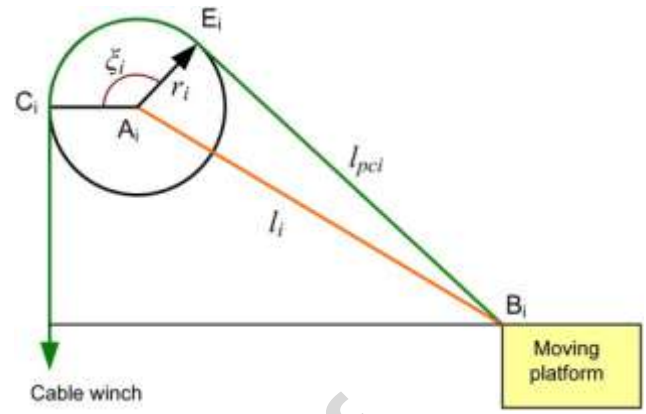

Figure 5 Pulley kinematics

was shown in Ref. [117] that it is possible to determine the lowest stable equilibrium state of suspended CDPRs with an arbitrary number of cables. An algorithm called branch-and-bound is used in this method to reduce the potential energy of the end-effector. A study of the kinematics of four mobile cranes transporting cable-borne payloads found that they operate in a collaborative manner [42]. To improve the precision of end-effector positioning, kinematic inaccuracies originating from machining, assembly, and operation were analysed. Also, in Ref. [43], three quadrotors carrying a payload were analysed from an inverse kinematics perspective. This configuration has an indefinite number of solutions to the IK problem. However, having specified the cable tensions, the solution to the inverse kinematics problem is found to be finite. The literature uses a variety of approaches to account for how the winding mechanism affects the dynamic behaviour of CDPRs [118]. Table 3 represents the kinematic and dynamic characteristics of different types of cables. It summarises the primary characteristics and computational approaches relevant to each type of cable in CDPRs.

Table 3 The kinematic and dynamic characteristics of different types of cables

Cable Type	*Kinematic Properties*	*Dynamic Properties*	*Methods and Models*
Non- flexible cables without mass	1. Simplified FK using straight-line model; 2. Positional variables are easily determined; 3. Pulley and shape factors are not considered.	1. Simplifies dynamic modelling; 2. Use of Newton-Euler and Lagrange methods for dynamics; 3. Suitable for rapid and deterministic solutions.	1. Newton-Euler method; 2. Lagrange method; 3. Straight-line model.
Non- flexible Cables with mass	1. Less accurate straight-line models over long distances; 2. Cables assumed to be flat or arc-shaped; 3. Catenary and parabolic models consider curvature.	1. Mass contributes significantly to system dynamics; 2. Consideration of pulley dynamics; 3. Adaptive Runge-Kutta algorithms used for PDEs.	1. Catenary model; 2. Parabolic model; 3. Adaptive Runge-Kutta algorithm.
Flexible Cables (Elastic)	1. Rate of cable length intensification is considered;	1. Linear or nonlinear stiffness behaviours;	1. Euler-Lagrange formulation; 2. Linear spring model.

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	2. Includes cable elasticity in kinematic modelling; 3. Elongation due to tension and stiffness.	2. Damping coefficient inversely related to cable length and elasticity; 3. Euler-Lagrange formulation for dynamics.	
Hefty and flexible Cables	1. Irvine sagging-cable model used to describe shape under gravity; 2. Kinematics influenced by mass and flexibility of cables; 3. Kinematic and force singularities analysed.	1. Dynamic stiffness and stability analysed; 2. Lumped-mass method for simplifying cable mass modelling; 3. Finite-difference method for ODE discretization; 4. Natural frequency analysis used for stiffness calculations.	1. Irvine sagging-cable model; 2. Lumped-mass method; 3. Finite-difference method; 4. Port-Hamiltonian method; 5. Natural frequency analysis.

In the context of cable guiding mechanisms, one vital component that demands attention is the cable outlet mechanism situated at the base of CDPRs. This mechanism exhibits diverse structures and can be categorized into four primary configurations: eyelet [119,120], single-pulley [90,121,122], double-pulley, and multi-pulley [123–125] configurations. The eyelet-type cable outlet mechanism represents a scenario where the cable's exit point remains fixed in relation to the robot's base. In this configuration, the cable's attachment point does not move as the robot operates. While this seems like an ideal setup, it introduces challenges related to friction and wear. The relative motion between the cable and the eyelet generates significant frictional forces, leading to the potential for cable wear and eventual breakage. This phenomenon can severely affect the reliability and practicality of CDPRs employing this design. The single-pulley type incorporates a single pulley through which the cable passes. This configuration allows for some degree of cable motion, as the pulley can rotate, facilitating cable movement. However, it is essential to consider factors like pulley friction and wear when employing this design, as excessive friction can affect the efficiency and longevity of the cable-driven system. The double-pulley design features two pulleys through which the cable traverses. This arrangement enables more freedom of cable movement compared to the single-pulley type. By having two pulleys, the cable can be guided more smoothly, reducing friction and wear. This design is particularly valuable for minimizing cable-related issues and enhancing the system's reliability. The multi-pulley type extends the concept of the double-pulley design by incorporating multiple auxiliary pulleys in the cable's path [125]. The auxiliary pulley acts as a mechanical leverage enhancer, optimizing the torque transmission characteristics within the CDPR system. This deliberate engineering choice is particularly beneficial when confronted with the challenges of friction at the swivelling axis, ensuring that the CDPR can smoothly and effectively execute swivelling motions. In Ref. [124], statics of 6-8 RRPM considered the effect of cable slipping. In contrast to the conventional methods of fixing the cable end to the pulley and winch, the study proposed a different approach by utilizing an endless cable or loop cable wound around an endless pulley and winch. The research introduces an interpretation of Euler-Eytelwein's formula through a graphical representation, wherein the non-slipping condition is visually represented as an area within the cable tension.

A crucial consideration in the transmission system of cable robots is compensating for variations in cable length due to pulley kinematics [110]. The dimensions of the guide pulley located at the cable exit point play a pivotal role in influencing the accuracy of solving the inverse kinematics problem. In scenarios where the anchor points at the moving platform are assumed to be fixed, the exit point at the base frame becomes dynamic, contingent on factors such as the size of the pulley, the position of the moving platform (MP), and the angle of cable contact on the pulley, as illustrated in Figure 5 [110]. The model accounts for the dynamic relationship between cable length, pulley radius, and the wrapping angle of the cable on the pulley [126]. The compensation for cable length l_{cpi} considering pulley effects is expressed mathematically as follows:

$$l_{cpi} = l_i + r_i.\xi_i \,, \tag{2}$$

where l_i is the length of the ith cable, r_i is the radius of the pulley associated with that cable, and ξ_i is the wrapping angle of the cable around the pulley. This equation represents the comprehensive model for cable length compensation, incorporating the influence of pulley kinematics, and is instrumental in refining the accuracy of the inverse kinematics calculations in cable robots. The compensation term $r_i.\xi_i$ reflects the additional length introduced due to the cable's interaction with the pulley,

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acknowledging the dynamic nature of the exit point at the base frame. In Ref. [127], the mobile platform was fitted with reflective pulleys in order to improve the robot's kinematics and dynamics. To account for their effects, these reflective pulleys must have the same diameter as those on the base platform. Based on Ref. [9], pulleys only have a substantial influence on the kinematics of compact CDPRs, and otherwise can be ignored. It was demonstrated in a study that a spatial IRPM with a five-bar mechanism increased loading capacity and decreased drive torques. Spatial and planar configured robots were studied in kinematic and dynamic analysis using geometrical concepts by taking into account the dynamics of pulleys and winches [118,128]. In Ref. [64], trajectory tracking was introduced for a planar 3-2 CRPM. To minimize the forward kinematic error, the closet set method was used to search for the position. Dynamic stability is evaluated analytically of a 3-3 IRPM lower limb rehabilitation robot [67]. CDPR kinematics and dynamics are used to define performance parameters such as position index, posture factor, and cable tension factor. The hybrid force-position-pose approach to CDPR is based on a weighted average method among three performance factors as an appraisal index. Then, according to the stability margin, stable workspace and robust workspace are defined.

In assessing the critical performance criterion of accuracy in CDPRs, a study compared performance enhancements achieved through different data-driven modeling and calibration approaches [129]. The focus extends to evaluating the effectiveness of CDPR models optimized through machine learning (ML) techniques, including linear regression, boosted regression trees, and neural networks. Well-optimized CDPR and ML models can significantly boost the practical accuracy of CDPR systems.

Despite the reduced computational complexity of the straight-line model, it is essential to acknowledge its limits in neglecting pulleys and cable geometric forms. Potential advancements in CDPR modelling should focus on including intricate geometric and dynamic elements to improve model precision without substantially increasing computing burden.

3.1.2 Non-Flexible Cables with Mass

A description of nonelastic cable models with mass, along with their effects on kinematics and dynamics, is provided in this section. In situations where the CDPR spans a considerable distance, and the weight of cables becomes a non-negligible factor, the conventional straight line model may suffer a decrease in accuracy. This decline in accuracy stems from the inherent mass, which contributes significantly to the overall dynamics of the system. For the analysis of CDPR kinematics and dynamics in the context of mass, two notable methods have been outlined in the literature. In the first method, cables are assumed to be flat in shape, while in the second approach, they are considered to be bulky, or arc shaped. Based on the first method, cables with varying masses and velocities were assumed to be straight lines [130]. Based on a Lagrangian method, dynamics expressions for a 4-3 planar IRPM were obtained. Based on the principle of virtual work, dynamic equations of motion for the FAST structure were developed, while simultaneously applying the Newton-Euler equation for a varying mass system to the cable [131]. Dynamic forces arising from a typical trajectory are contributed by the limb's varying length and velocity cable. It was proposed that cables be modelled as cylinders in Ref. [132]. An analysis of motion by employing the spherical bushing joint on ADAMS software was conducted for a 7–6 CRPM. An alternative second method was presented in Ref. [133], where the structure of cable is defined by a hyperbolic function. In contrast, the catenary model offers high accuracy by capturing the inherent curvature of a hanging cable, providing a more realistic representation. However, its nonlinear equations necessitate iterative calculations, making it computationally intensive and less suitable for real-time control applications. The parabolic model serves as a computationally more tractable approximation of the catenary model. While it sacrifices some accuracy compared to the catenary model, the parabolic model offers a more straightforward solution process. This simplicity in solving equations makes the parabolic model more amenable to real-time control applications, where computational efficiency is crucial. A similar approach, which incorporates simplified static analysis, has been proposed to address the complexities associated with CDPRs. This methodology involves describing externally applied wrenches and forces in the cables through a linear relationship, as elucidated in Ref. [134]. The linear relationship facilitates a more straightforward representation of the system's static equilibrium, allowing for efficient analysis of external loads and cable forces. For scenarios involving a hefty configuration of cables, a detailed analysis was conducted by discretizing the cables into 'N' elastic segments and connecting them at nodes. This discretization approach enabled the derivation of dynamic equations, offering a thorough understanding of the interactions within the cable-driven system [135]. The application of adaptive Runge–Kutta algorithms proved instrumental in solving the resulting partial differential equations, employing lumped-mass methods to capture the distributed nature of cable mass and dynamics. Expanding the scope of analysis, considerations were extended to

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include not only cable mass but also the dynamics of pulleys in the kineto-static analysis, as documented in Ref. [11]. This comprehensive approach takes into account the influence of both cable and pulley dynamics, providing a more holistic understanding of the system's behaviour. Furthermore, addressing the direct geometric–static problem of spatial IRPMs, an innovative interval-analysis algorithm was employed [136]. This algorithm, while considering factors such as slack in the cables and the mass of the end-effector, facilitates the determination of all possible equilibrium poses. The incorporation of slack in the cables and the end-effector mass enhances the algorithm's ability to identify realistic and feasible equilibrium configurations, thereby contributing to a more accurate representation of the system's static behaviour.

While the catenary and parabolic models offer more accurate depictions of cable forms, their high processing requirements compromise their suitability for real-time applications. Hence, adopting a well-balanced strategy that uses machine learning methods to estimate cable behaviour in real-time could provide the optimal combination of precision and efficiency.

3.1.3 Non-Flexible Cables with Mass

Vibrations of cables in transverse directions can be neglected when compared to vibrations in axial directions. A 7–6 CDPR was studied in Ref. [137] in terms of its axial and transversal vibrations. In terms of vibration amplitude, transversal vibration accounts for only 1.4%. In order to design a cable manipulator that can be stiff enough for a variety of applications, the natural frequencies of the multi-body system were analysed. For CDPRs using elastic cables, cables are generally modelled as axial springs. In such case, the rate of cable length Intensification ΔL is define as:

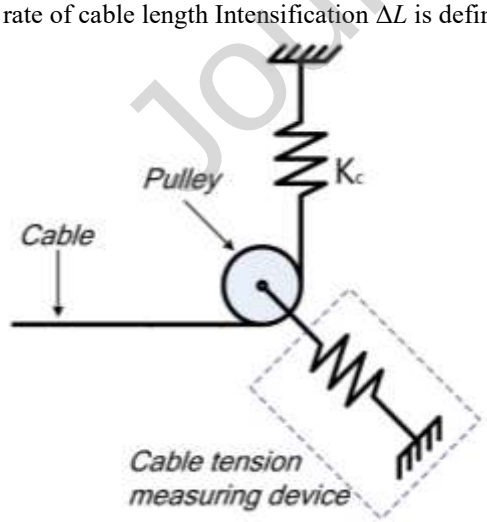

Figure 6 Typical Elastic model of cable transmission system

$$\Delta L_i = l_i - l_{oi} = K_{ci} T_{ci}, \tag{3}$$

where l_{oi} refers to the length of the cable when deformation of cable transmission mechanism is taken into consideration, ΔL_i is the elongation of the *i*th cable calculated from the stiffness K_{ci} and cable tension T_{ci} respectively. There is a possibility that cable stiffness can exhibit linear or nonlinear behaviour, but the majority of the literature is focused on linear stiffness, i.e., $K_{ci} = EA/l_i$, where E represents cable's young's modulus while A represents the cross-sectional area, and l_i specifies cable length when at a standstill.

In Ref. [15], introduced the inverse and forward kinematics of a 7–6 IRPM featuring elastic cables, assuming that tensions must remain below a predetermined threshold to prevent cable breakage. The calculation of potential energies and kinetic energies for the end-effectors and elastic cables was carried out using the Euler-Lagrange formulation [138]. There is an inverse relationship between the damping coefficient with cable length and the elasticity of the cables when considered as straight lines [139,140]. In order to derive the dynamics equation, the Euler–Lagrange formulation approach was adopted. For an 8-6 RRPM, a cable tension measuring device has been incorporated. To streamline the complexity of the calculation model, both the elastic properties of the cable and the tension measuring mechanism are conceptualized using a linear spring model, as illustrated in Figure 6 [110].

The underlying challenge is in establishing a balance between the requirement for precise elastic modelling and the computational requirements it entails. An effective approach could involve the creation of hybrid models that employ simplified elastic equations during low-stress activities and transition to more intricate models when required.

3.1.4 Hefty and Flexible Cables

Hefty flexible cables are a common feature of CDPRs and other cable-based systems. In these systems, the cables themselves are not ideal, massless, and perfectly non-elastic components, but rather have some physical properties that affect their behaviour and performance. The mass of the cable refers to the fact that the cables themselves have a certain weight and therefore will exhibit a degree of inertia when subjected to external forces. This means that the motion of the cables will be affected by the acceleration and deceleration of the robot and can, therefore, impact the overall behaviour of the system. The elasticity of the cables refers to the fact that they will stretch and deform when subjected to tension or compression. This property is essential to the functioning of many cable-driven parallel

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robots, as it allows the cables to transmit forces and motion between different parts of the system. Modelling cables with mass and elasticity can be a complex task, as it involves accounting for the additional degrees of freedom introduced by these physical properties. However, it is an essential step in accurately modelling and controlling cable-driven parallel robots, and many researchers have developed models and algorithms specifically tailored to this task.

Kinematic and dynamic equations are extracted from the Irvine sagging-cable model by taking into account both flexibility and cable mass [141–143]. A sagging cable model, also known as the Irvine sagging-model is a mathematical representation employed for describing the shape of a cable under the influence of gravity. This model is often used in the design and analysis of cable-based systems such as bridges, cranes, and cable-driven parallel robots. There are different formulations to describe such model and one of them is:

$$x_b = F_x\left(\frac{l}{EA} + \frac{\sinh^{-1}\left(\frac{F_z}{F_x}\right) - \sinh^{-1}\left(\frac{F_z - \mu g l}{F_x}\right)}{\mu g}\right), \quad (4)$$

$$z_b = F_x\left(\frac{F_z l}{EA} + \frac{\mu g l}{2} + \frac{\sqrt{F_x^{\,2} + F_z^{\,2}} - \sqrt{F_x^{\,2} + \left(F_z - \mu g l\right)^2}}{\mu g}\right), \quad (5)$$

where A and E is same as defined in the above section where as μ, F_x, F_z are the linear density of the cable material, horizontal and vertical forces respectively.

In the context of the FAST telescope, the Irvine sagging-cable model was used to set up the cable configuration. This involved determining the length, tension, and orientation of the cables that support the telescope dish, based on the physical properties of the cables and the gravitational forces acting on the cables [13,133] . In order to enhance the controllability of the system in real time, the kinematic and force singularities were analysed. Static cable displacement was evaluated in Ref. [106] as a basis for inverse kinematics and resistance to elongation calculations for CDPRs. A numerical continuation scheme was used to solve the forward kinematics of spatial IRPMs [141]. The forward kinematics was calculated for nondeformable cables. An analysis of stiffness and stability was performed based on linearized Irvine's sagging cable model [16]. Inverse kinematics of spatial CDPRs were solved using the cable tensions.

The Port-Hamiltonian method is a mathematical framework used for modelling and analysing dynamic systems. In the context of CDPRs, this method was employed to construct dynamic equations based on the total kinetic and potential energies of the mobile platform and cables [105]. The lumped-mass method is a technique used to simplify the modelling of cables with mass by approximating them as a series of discrete masses connected by springs. The finite-difference method is a numerical technique used to discretize differential equations. In the context of CDPRs, this method was utilized to discretize the ODEs obtained from the lumped-mass method, enabling numerical calculations of the system's behaviour [144]. In addition to the Irvine sagging-cable model mentioned earlier, a spring model was also used to describe the behaviour of cables in some CDPR studies. System equations were derived from the Newton-Euler algorithm in this case [128]. To calculate the static stiffness of the cables, a hefty elastic cable model illustrated in Figure 7 can be used based on the variation of the end-effector's error in position [145]. This involves first applying a known displacement to the end-effector and measuring the resulting change in cable tension [145]. A natural frequency analysis is a type of dynamic analysis that is commonly used to determine the dynamic stiffness of a structure, including CDPRs. The natural frequency of a structure is the frequency at which it will vibrate when excited by an external force. The dynamic stiffness of the CDPR can be determined by analysing its response to small amplitude vibrations at different frequencies. In Refs. [80] and [146], numerical methods were used to calculate the stiffness matrix for spatial suspended CDPRs. These calculations involved modelling the CDPR as a system of equations that relate the cable tensions to the robot's motion. The stiffness matrix was then calculated from the resulting equations using numerical methods. For design purposes, a homogenization of this stiffness matrix has been presented in Ref. [146]. This involves simplifying the stiffness matrix by assuming that the cables have uniform properties, and that the robot's motion is constrained to a particular direction. This simplification allows for easier analysis of CDPRs performance and optimization of the system's design. The forward and inverse kinetostatic problem of CDPRs involves calculating the position, velocity, acceleration, and forces of the robot's end-effector given its cable lengths and vice versa. This problem is highly nonlinear and computationally complex, and numerical methods are typically used to solve it. However, these methods are often unreliable in real-time scenarios because they are heavily dependent on initial values, iterative, unbound by computations, and slow to converge. To address this issue,

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the paper proposes using DNNs to solve the kinetostatic problem. DNNs are a type of machine learning algorithm that can learn complex mappings between inputs and outputs. By training a DNN on a large dataset of cable lengths and corresponding end-effector positions, velocities, accelerations, and forces, the network can learn to predict these values for any given set of cable lengths [147].

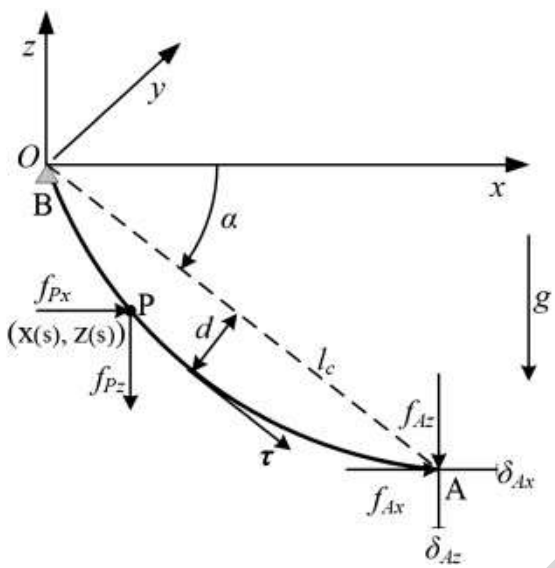

Figure 7 Diagram of a sagging cable

Considering the advancements made in modelling and computational techniques, there remain a few unresolved issues, particularly handling the real-time dynamics of elastic and hefty cables under different load conditions. Moreover, the interaction between cable mass and system stability requires further investigation to enhance the accuracy of CDPR operations in complex scenarios. Though conventional models such as straight-line and catenary models are computationally efficient, they frequently lack to accurately represent the intricate characteristics of heavy and elastic cables. From our perspective, incorporating data-driven methods with classical mechanics in hybrid approaches could offer a stronger foundation for CDPR analysis. Integration of the Irvine sagging-cable model with real-time data from sensors has the potential to greatly enhance the precision of kinematic predictions.

3.2 Design and Workspace

The reachable workspace is defined as the volume of space within which every point can be reached by the end-effector in at least one orientation. This means that the robot can reach any point within this volume by moving its end-effector, while keeping it oriented in a way that allows it to reach that point. The reachable workspace is a fundamental concept in robotics, as it helps to define the capabilities of a robot in terms of the tasks it can perform and the environments it can operate in. By analysing the reachable workspace of a robot, it is possible to identify any limitations or potential areas for improvement in its design, as well as to optimize its operation for specific tasks and environments. The controllable or wrench-closure workspace (WCW) is the region of the robot's workspace where it is possible to achieve a desired wrench output, meaning that the tensions in the cables are greater than zero [148]. In the configuration space of the robot, this region corresponds to all possible position of the end-effector. The wrench-feasible workspace (WFW) is the region of the robot's workspace where it is possible to achieve a desired wrench output, while also ensuring that the tensions in the cables remain within a positive lower and upper bound. In this region, the robot is able to resist any external wrenches that are applied to it.

The exploration of CDPRs has led to diverse applications and analytical investigations, revealing insights into their workspaces and collision-free configurations. In a study focusing on a 2-link serial CDPR with repetitive workspace was presented [95]. The 2-2 IRPM was utilized for the spray-painting task, and the repetitive workspace allowed the platform to be shifted several times. Analysing geometric considerations for serial manipulators, researchers delved into the impact of anchor point locations on workspace configurations. Utilizing conic sections, such as parabolic and elliptic curves, the WCW of a 4–3 CRPM was determined for different mobile platform angles [148]. CRPM-configured planar and spatial CDPRs WCW was determined using a variant of Bland's pivot rule [149], revealing that converting equilibrium inequalities into equations offers an effective calculation of workspace compared to numerical approaches. A study focused on prestressed cables in the context of 4-3 CRPM introduced the concept of prestress stable WCW, identifying a subset of the WCW that increases stiffness with prestress augmentation [150]. This investigation contributes to optimizing the structural stability and performance of CDPRs through cable prestressing. Furthermore, interference free workspace analysis for a modularly constraint robots in cluttered environments was explored in Ref. [26]. The authors employed various techniques, including topological constraints, critical support lines, and convex hull mapping. Critical support lines, determined by connecting furthest points of consecutive obstructions, delineate regions in the workspace susceptible to collisions.

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Convex hull mapping involves creating a polygon surrounding all obstacles, defining the boundary of the collision-free workspace. Topological constraints, based on homotopy principles, ensure the robot's navigational safety by allowing continuous deformation of paths without encountering obstacles. This multifaceted approach offers a comprehensive understanding of the collision-free workspace for CDPRs in complex environments, emphasizing the significance of geometric considerations and innovative computational techniques in enhancing their operational safety and efficiency.

In Refs. [151,152] the effects of various parameters on the workspace geometry of spatial IRPMs were investigated, including the sizes of the mobile and base frame as well as the end-effector's rotation angle. Observations reveal that the largest workspace is obtained when the end-effector's degree of orientation is zero. For spatial CDPRs with more than six cables, the boundaries of the WCW were defined in Ref. [153]. The WCW for such robots is typically composed of cubic surfaces, such as the WCW for Gough-Stewart robots. An analytically determined dynamic workspace was obtained for the end-effector based on a specified set of accelerations in Ref. [154]. In a previous study, the end-effector's inertial, external wrench, and centrifugal and Coriolis matrices were taken into consideration to improve the workspace of a 4-6 IRPM [155]. The inclusion of these factors is important because they can have a significant impact on the overall performance and capabilities of the robot. For example, the inertia of the mobile platform can affect the stability and manoeuvrability of the manipulator, while external wrenches can cause unwanted movements or vibrations. In spite of this, the improved workspace was observed to be approximately 32% less spacious than the general workspace. It is always the tension status of a CDPR's driving cables that identifies and characterizes its workspace. Therefore, evaluating the quality of the workspace can be determined by looking at the distribution of tension among the cables. An optimal tension factor value for determining an optimal workspace was established [156]. The tension factor is defined as the ratio of the minimum tension of the cables to the maximum tension of the cables. A value close to zero indicates that the end-effector is located near the boundary of the workspace. Similarly, when the factor approaches one, the end-effector will be located farther away from the region defined by the workspace boundary. A 12-3 RRPM with translational contour crafting was shown with four upper cables to primarily support the weight of the end-effector, while the rest of the lower cables drive the required translation motion [77]. A maximum structure size was estimated by using cable tensions for this robot under a wide range of loading conditions. A study introduces a set of newly defined fundamental wrenches, which are used to determine the worst-case scenario for the forces and torques that can be exerted on the moving platform of the robot [157]. The method can identify the critical points of the robot's workspace where the forces and torques are the highest. The study also notes that the proposed method is applicable to any CDPR with redundant actuation, as long as the Jacobian matrix of the cable manipulator is of full rank. This means that the method can be used for a wide range of cable-driven robots, regardless of their specific design or configuration. Analysis of workspaces and studies of force singularities were conducted based on the mass of cables. A CDPR with a large workspace, like FAST [51], uses such cable models. Due to the fact that all cable tensions pass through a common straight line, the cabin may not be able to balance when a singularity occurs. A redundant cable was recommended for handling force singularities and improving the workspace to address this problem. The static or quasi-static problem of the FAST was resolved by an analytical solution method [158]. This method takes less time to compute than any iterative methods, making it ideal for control in real-time. A study that focuses on the multiplicity of kineto-static solutions for redundant CDPRs presented three different selection criteria for determining a unique kineto-static solution of RRPMs [159]. The three selection criteria presented in the paper are as follows.

1. Minimizing the tension in the cables: This criterion aims to minimize the overall tension in the cables of the robot, which can help to reduce wear and tear on the cables and increase the robot's lifespan.
2. Optimizing the deviation between cable tension and mean tension: This indicator aims to minimize the differences in tension between the cables, which can help to ensure that the robot moves smoothly and accurately.
3. Maximizing the tension in the cable: This criterion aims to maximize the tension in the cables, which can help to ensure that the robot can handle larger loads and operate more efficiently.

The paper notes that the appropriate selection criterion will depend on the specific requirements of the application. For example, if cable wear and tear is a major concern, the first criterion may be the most appropriate. If accuracy is the top priority, the second criterion may be the best choice. If the robot needs to handle heavy loads, the third criterion may be most suitable.

The literature describes two approaches to optimizing CDPR workspaces. An initial approach examines the effects of different configurations of cable and motor. The other

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entails integrating springs into their design. By using numerical methods, the first method was applied to evaluate the WCW of WIRO 6.1 and WIRO 4.3, which is a 7–6 CRPM [160]. This method was similarly applied to evaluate the WCW of a 9–6 RRPM named WIRO-6.3 [161]. The outcome indicated that WIRO-6.3 offered the most workspace. The first approach was also utilised in another study to compare three different 7–6 CRPMs cable configurations. A performance index was developed using the maximum permissible distance between the geometric centre and the mass centre of a moving frame [162]. In order to maximize load carrying capacity and workspace, motors and winches were optimally placed. An 8–6 RRPM workspace was improved with the help of Wire Centre software [27]. It was demonstrated that different cable configurations enhanced rotational workspace considerably. Moving the platform across the workspace requires fewer forces to be applied with the use of this mechanism. As a result, the workspace geometry was maintained by ensuring minimum tension in the cables. Ref. [50] investigated end-effector geometry in pick-and-place problems, using a genetic algorithm to maximize the space required for overall orientation WFW. By using this method, the robot actuator's maximum tension is not excessively increased, so the method is effective in terms of optimising workspace.

A second approach was used in Ref. [163], in which a constrained robot's moving platform was integrated with an articulated gripper. By grabbing an external structure, the gripper extends the workspace boundaries, allowing the end-effector to reach beyond its standard configuration space. A study reviewed the effects of adding springs to the planar and spatial CRPMs [37]. The number and location of these springs were varied in order to examine their effect on the moving platform wrench. An optimization scheme based on quadratic programming was applied to calculate the desirable spring parameters with lower and upper limits on cable tension in order to maximize WFW. It was shown that two cables could be pulled using just one actuator by using a differential mechanism [36]. In convex theory and linear algebra, the workspaces of CRPMs with and without this mechanism were investigated. It was demonstrated that a cable loop structure can provide unlimited and singularity-free orientation workspaces for planar CDPRs [164]. It was investigated that while the cable-loop structure can provide many benefits, it can also result in occasional rotational motions of the end-effector, which are known as parasitic inclinations. To address this issue, the authors developed a model to assess and predict the occurrence of these parasitic inclinations. The model developed can help to optimize the design of CDPRs with cable-loop structures by predicting the occurrence of parasitic inclinations and minimizing their impact on the robot's performance. This can help improve the accuracy and reliability of the robot, particularly in applications where precise orientation is critical. In a study it was investigated that the use of bar linkages and springs can help to reduce the forces required to move the end-effector across the workspace of the CDPR [165]. This mechanism can also help to maintain a minimum desirable tension in the cables, which is important for conserving the geometry of the workspace. A study introduced an 8-6 CDPR featuring movable anchor points, adding a dynamic dimension to the traditional design [166]. In this approach, the anchor point for each driving cable has the capability to move along a track integrated into the fixed frame (Figure 8(a)) revealing a substantial enhancement in the wrench-feasible workspace as illustrated in Figure 8(b) [166]. The study not only establishes the kinematics and dynamics of these mobile anchor points but also introduces a collision detection method and an obstacle avoidance algorithm to enhance its adaptability and operational safety. The proposed obstacle avoidance method ensures that the CDPR can intelligently navigate through its environment, dynamically adjusting its cable configurations to avoid collisions while adhering to a predefined trajectory. The study employs the creation of polygonal envelopes for both the moving platform and obstacles. The moving platform is encapsulated within a convex polyhedron envelope, providing a simplified representation for collision assessment. The collision detection algorithm utilized in this research is the efficient OpenGJK fast detection algorithm, ensuring rapid and accurate identification of collisions with obstacles. For static obstacles, collision determination involves calculating the shortest distance between the cable and the obstacle, allowing for a precise assessment of potential collisions. Additionally, when dealing with moving obstacles, collision determination is accomplished by evaluating whether the sweeping space of the axis-aligned bounding box (AABB) envelope of the obstacle intersects with the cables. This comprehensive collision detection strategy combines various techniques tailored to different scenarios, offering a robust methodology for ensuring the obstacle avoidance capability of the CDPR. With a given trajectory, Figure 8(c) represents the areas of realization for obstacle avoidance [166]. For an IRPM, assessing the robot's workspace becomes non-trivial, despite its critical importance in performance analysis. A novel algorithm for computing the reachable static workspace of generic underactuated CDPRs is demonstrated on robots with 3, 4, and 5 cables, defining it as the set of end-effector positions achievable with stable

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static equilibrium and positive, bounded cable tensions [167]. The algorithm addresses a geometrically static problem, seeking orientations that maintain the desired tension distribution for a given EE position. An analytical formulation of the Jacobian facilitates quick computation. Results reveal that cable tension limits are the primary factors restricting the reachable static workspace for any architecture. Moreover, the algorithm is effectively implemented on a 4-6 IRPM to compute continuous feasible set points for trajectory planning.

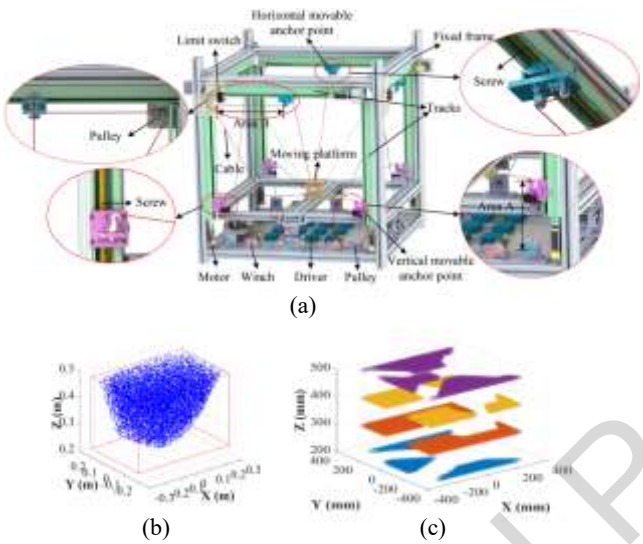

Figure 8. (a) A CDPR with a reconfigurable anchor point for collision free path, (b) wrench-feasible workspace of the reconfigured CDPR, (c) Areas realized for obstacle avoidance with a given trajectory

Current analyses often overlook the dynamic aspects of workspace evaluation, with most studies focusing on static configurations and neglecting real-time adaptability. Research on the impact of varying anchor point locations and the integration of advanced computational methods for workspace optimization is limited. The interplay between cable tension and workspace boundaries under different load conditions also requires further exploration. Existing methods for evaluating workspace quality are predominantly static and neglect to account for real-time adaptability or the influence of external factors, such as environmental changes. Additionally, research on integrating spring mechanisms with CDPRs is still in its early stages, and the long-term effects of these mechanical adjustments on workspace stability and performance are not well understood. The effectiveness of current obstacle avoidance algorithms in dynamic and unpredictable environments needs further validation, and there is a lack of comprehensive findings that integrate collision detection with real-time workspace adaptation.

3.3 CDPRs Path Planning

A crucial component of the design and operation of CDPR is path planning. Path planning can be difficult for CDPRs because of the complexity of their workspace and the possibility of problems such cable collisions and singularities. To avoid cable sagging during operation, CDPR trajectory planning must remain within acceptable bounds for cables of different lengths and tensions [168]. It is easier to plan the trajectory for CRPMs and RRPMs since all DOF on the moving platform can be controlled. The trajectory planning for IRPMs is complicated because no WCW exists. There has been extensive research on path planning for CDPRs, with many different approaches proposed and evaluated. The choice of approach will depend on the specific requirements of the application, such as the desired level of precision, the complexity of the workspace, and the constraints on the robot's motion.

3.3.1 Path Planning of Translational CDPRs

Translational CDPRs path planning is the process of designing trajectories that allow the robot to follow a predetermined path while ensuring stability and precision. Robotic systems of such nature are commonly employed in situations that need accurate translational movement. Their algorithms must consider constraints such as velocity, acceleration, and external forces. The optimisation of the robot's movements and adherence to limitations during activities such as graffiti removal, limb rehabilitation, and other industrial applications have been addressed by several proposed methodologies.

To constrain the out-of-plane motion of a 2-2 IRPM, a method was introduced, involving the attachment of a two-link serial robot to a moving platform. This approach helps to reduce the unwanted motion of the moving platform and improve the accuracy and stability of the robot [53]. A numerical approach was employed to calculate the minimum-time trajectory that adheres to velocity and acceleration constraints for a specified path. To simplify the method implementation in an industrial robot controller, the authors employed the trapezoidal-velocity profile trajectories. This type of trajectory is widely used in industry and is easy to implement in real-time control systems. By using this type of trajectory, the authors were able to optimize the motion of the robot while ensuring that the velocity and acceleration constraints were met. For removing graffiti from highway signs, the 2-2 IRPM was proposed [96]. To create optimal force over the entire

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surface of the signpost, rotary arms were installed at the upper ends of the robot. Graffiti and non-graffiti were separated by computer imaging into 20 sections wide by 20 sections high in the workspace. Each section's coordinates were then interconnected to generate the robot's path. This approach allowed the robot to efficiently cover the entire surface of the signpost and remove the graffiti with high precision and accuracy. For lower limb rehabilitation, a reconfigurable 4–3 CRPM was developed [169]. A fixed platform is equipped with cable attachment points that can be reconfigured during limb motion in order to balance the external forces. To increase the WCW of this mechanism, the actuation configurations were changed. A range of methods is used for CRPMs and RRPMs to plan the minimum-time trajectory. An analysis of the effects of cable configurations, switching points, and cable numbers was used to plan a minimum time trajectory for a 4–3 CRPM [170]. An approach to minimum-time path planning for redundant and non-redundant cable robots was presented [113]. The limits on cable tensions and velocities can be transformed into bounds on accelerations and velocities along a path by reducing the equations of motion to a system of second-order differential equations with path parameters 's'. An analytical approach aimed at planning dynamic point-to-point trajectory for 6-3 RRPM was presented in a study. From a static condition, the robot is guided through a sequence of dynamic manoeuvres that culminate in precise launch of an object to a predetermined target. The study uses the smooth and controllable characteristics of a second-order Bezier curve. In order for the cable robot to maintain stringent kinematic constraints, it orchestrates the motion through this trajectory [171]. Another study followed a similar approach for the reference trajectory of 4-3 CRPM [172].

The current research landscape in translational CDPR path planning is dominated by application-specific solutions. While these approaches have yielded favourable outcomes in their respective domains, there is a need for a more broader framework that can be adapted to a wider variety of tasks. Furthermore, integrating machine learning techniques to predict and adapt to dynamic changes in the environment could enhance the potential to improve the efficiency of path planning in real-time applications.

3.3.2 Path Planning of Spatial CDPRs

Path planning for spatial CDPRs focuses on generating trajectories between two or more points while maintaining certain constraints, such as acceleration continuity and zero velocity at the target points. It entails selecting a path that avoids obstacles while satisfying the robot's kinematic and dynamic constraints. In point-to-point CDPR path planning, several different approaches are used. An approach to connecting consecutive target points with zero instantaneous velocity is to use hypocycloid curves for generating trajectories in CDPRs. A hypocycloid curve allows target points to be located outside the robot's static workspace, thereby extending the robot's effective workspace. In Ref. [173], this approach was successfully applied. A 3–3 CDPR trajectory expression was derived using trigonometric operations in Ref. [174]. In this method, trigonometric functions are used to determine the robot's position, velocity, and acceleration. Indirect paths were selected by specifying viable points between two consecutive points. In dynamic trajectory planning of a spatial 3-3 IRPM for point-to-point movements and periodic motions, as presented in Ref. [175], one of the approaches used to design point-to-point trajectories is by analysing the geometric characteristics of positive cable tension constraints in the $s - \ddot{s}$ plane. In this approach, the $s - \ddot{s}$ plane represents the path parameters that is position and acceleration states. A camera equipped IRPM was presented with another approach to path planning [86]. By gathering feedback measurements from the camera of object's current positions and velocities, the intended position and velocity of the camera are calculated. With parallelogram cable configurations, robot movement can be controlled more effectively. In terms of workspace and trajectory, this provides greater flexibility. The robot's motion is made less uncertain by constraining the cables into parallelograms, which eliminates any slack in the cables. The feasible dynamics of a 6–6 IRPM were changed with parallelogram configurations of cables [176]. There are two parallel cables of equal length in each parallelogram. However, the robot generates three-dimensional translational motion in a similar orientation to the moving platform. In contrast to CRPMs or RRPMs, IRPMs involve fewer cables in their structure. This enables the payload to have additional degrees of freedom as well as exhibit oscillations or swaying motions. Consequently, this design implies that only a subset of the end-effector DOF can be effectively controlled for planning purposes, and that the end-effector can move freely even when the actuators are inactive. By means of a zero-vibration input shaping scheme, trajectory planning was implemented for 4-6 IRPMs to eliminate unwanted oscillations [177]. In order to calculate the natural frequency of planar IRPMs, motion in three-dimensional space was mirrored onto two perpendicular vertical planes. Underactuated CDPRs with rest-to-rest (RTR) motions were investigated in a study with prescribed motion times and path geometry [178]. Due to the constraint deficiency on the end-effector motions, arbitrary trajectory requirements for a

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suitable subset of end-effector coordinates can lead to system instability and the inability to bring the system to rest within the prescribed time. To address this, the study proposes a novel trajectory-planning technique based on solving a boundary value problem to ensure accurate path tracking and stable static pose transitions. The boundary conditions are defined as:

$$x(0) = \begin{bmatrix} q_u(q_a(0)) \\ 0 \end{bmatrix} := x_0 \,, \qquad x(T) = \begin{bmatrix} q_u(q_a(T)) \\ 0 \end{bmatrix} := x_T \,, \tag{6}$$

where $'q_u'$ and $'q_a'$ represent unactuated and actuated coordinates respectively are the stable equilibrium configuration of the system. An additional scalar parameters called free parameters are incorporated, so that $q_a = q_a(\mathcal{K}, t)$. The boundary value problem becomes:

$$\begin{cases} \dot{x} = f\big(x(\mathcal{K}, t)\big), q_a(\mathcal{K}, t), \dot{q}_a(\mathcal{K}, t), \ddot{q}_a(\mathcal{K}, t), \\ \qquad x(0) = x_0, \qquad x(T) = x_T. \end{cases} \tag{7}$$

Here '$\dot{x}$' represents the system state based on system dynamics. The effectiveness of the method is demonstrated through trajectory planning for underactuated 3-6 IRPM, validated experimentally. The work also addresses the complexities associated with solving the boundary value problem and introduces a heuristic iterative approach reminiscent of the classic iterative shooting method for determining essential free parameters, such as $'\mathcal{K}'$, which are crucial for the planning process. Unlike conventional methods such as the bvp4c and bvp5c routines, which are finite-difference codes utilizing a collocation formula, this iterative approach offers a practical alternative. These standard algorithms necessitate a well-structured setup to function efficiently and produce solutions within acceptable tolerances. The study effectively compares RTR and Standard (STD) trajectories. STD trajectories exhibit residual oscillations, attributed to dissipative effects and minor end-effector coordinate drift caused by cable oscillations. Figure 9 provides a comprehensive visual representation of the position and orientation trajectories of RTR and STD trajectories. Figure 9(a) through (b) depict these trajectories at various stages, capturing their evolution over time. Subsequently, in Figures 9(c) and (d), an enhanced view is presented, focusing specifically between the transition states [179]. This detailed depiction allows for a closer examination of the trajectory characteristics during this critical phase, providing valuable insights into the behaviour of the system. Nonetheless, RTR trajectories demonstrate stable transitions with limited oscillations, particularly evident in subsequent transitions, showcasing the efficacy of the proposed approach.

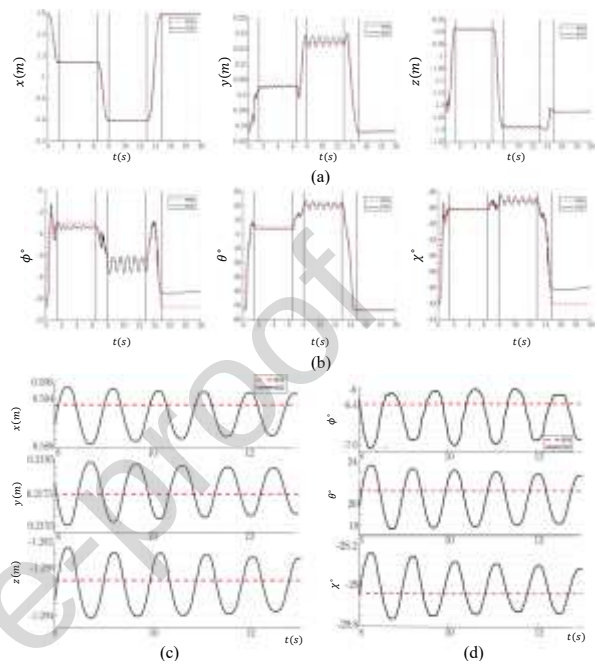

Figure 9 Pose of end-effector for RTR and STD trajectories. A vertical dashed line delineates set point transitions. (a) Position, (b) Euler Angles, (c) Enhanced view of most critical measured oscillations along reference position, (d) Enhanced view of most critical measured oscillations along Euler angles

A study introduced a similar methodology for computing the natural oscillation frequencies of an IRPM, addressing the challenges posed by the under constrained nature of the end-effector [179]. By linearizing the system's internal dynamics around equilibrium configurations, the proposed approach enables the determination of end-effector oscillation frequencies for CDPRs suspended by 2 to 5 cables. The described procedure was systematically implemented across three distinct scenarios to comprehensively evaluate its efficacy. Initially, it was applied to 30 equilibrium configurations wherein the platform was constrained by four cables. Subsequently, an additional set of 30 configurations was examined, wherein only cables 1 through 3 were affixed to the platform. In addition, 30 configurations were analysed, focusing on scenarios where the platform was solely suspended by cables 1 and 3. These configurations are illustrated in Figure 10(a) [179]. This comprehensive approach enabled a thorough exploration of the procedure's performance across various cable configurations, providing valuable insights

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into its versatility and effectiveness under different conditions. Figure 10(b) illustrates the experimental natural frequency $'f_j'$ representing computed oscillation frequencies whereas ' f_j^* ' represents experimental oscillation frequencies [179].

Detailed investigations into the kinematics, dynamics, stability, and stiffness of the robot's free motion validate the method through experiments conducted on 6-DoF models actuated by 2, 3, and 4 cables. Furthermore, the study demonstrates the practical relevance of this modelling strategy by applying it to trajectory planning for a 6-4 IRPM using a frequency-based method, showcasing reduced end-effector residual oscillations compared to traditional approaches. The approach underscores the effectiveness of the methodology, with experimental results showing oscillation frequencies deviating from the model by less than 6%.

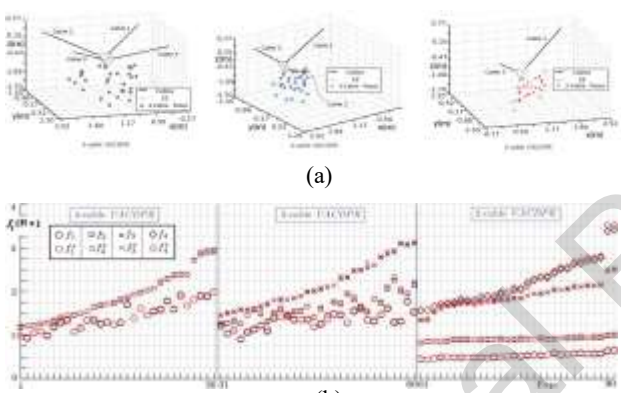

Figure 11 (a) Layout of experimental configuration, (b) Computed and Measured oscillation frequencies for CDPR with 2,3, and 4 cables [179]

In a study, a generalised methodology was proposed for determining IRPM equilibrium configurations at assigned end-effector positions. To achieve static equilibrium while considering cable tension constraints, this method solves an optimization problem to determine cable length, tension, and end-effector orientation [180]. The approach facilitates generating vibration suppression trajectories and obtaining cable lengths for a given end-effector position, even without an orientation trajectory. The input shaper is also designed using natural frequencies to enhance trajectory tracking performance and suppress oscillations. For a 4 and 5 cable configured IRPM in 3D space the number of equations is less than number of unknowns and there exists infinite solutions to the static and kinematic constraints. In order to degerming the unique feasible solution a numerical procedure based on convex analysis and optimisation with constraints is applied. The optimization problem is formulated by defining the objective function as the 2-norm of statics, while ensuring that all cables remain taut and that there is no twisting or interference among them. The equality constraint is defined to reflect kinematic constraints whereas inequality constraints are defined by minimum tension constraints and bounded range to prevent cable sagging and avoid interference respectively. An interior-point algorithm is used to determine optimal solution. Furthermore, two input shapers are designed for analysis are zero-vibration (ZV) method and zero-vibration derivative (ZVD) method. However, the performance in vibration suppression of the input shaped trajectories is compared for different angular velocities which are constant angular velocity (CV), trapezoidal angular velocity (TV), ZV and ZVD. The performance in vibration suppression is evaluated on two circular paths and their orientation angles at equilibrium are illustrated in Figure 11(a) whereas Figure

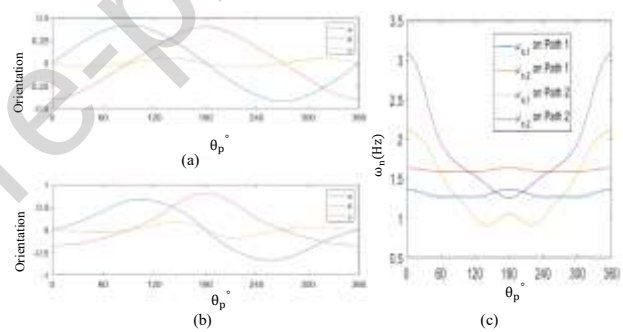

Figure 10 (a) The orientation angles at equilibrium along horizontal circular path, (b) The orientation angles at equilibrium along tilted circular path, (c) Natural frequencies along the path

11(b) represents the natural frequencies along the path [180]. For motion along the tilted circular, large changes in the orientation angles and natural frequencies are caused. To assess the effectiveness of an equilibrium-based trajectories along the trajectory where significant changes occur, the behaviour of the end-effector was observed. The method mitigated both the residual oscillations induced during circular motion as well as the transitional oscillations induced during circular motion across all cases.

In the case of fully or redundantly actuated robots, traditional performance indices may not directly apply to IRPM manipulators. To address this concern, a novel index called "maximum tension variation under a cable displacement error" is proposed specifically tailored to suspended and underactuated CDPR [181]. This index evaluates cable-tension sensitivity to cable-length variation and aims to assess the feasibility of achieving geometric

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static equilibrium under controlled cable-length errors. Its applicability was demonstrated on a 4-6 IRPM, highlighting orientations that minimize and maximize the index when assigning the end-effector position. Furthermore, its relevance was illustrated by emphasizing configurations where errors in cable length pose the highest risk of tension loss. This index serves as a valuable tool for characterizing and optimizing the reachable workspace of IRPMs. An equivalent spring model of the cables was used to plan the trajectory of a 6-6 IRPM [182]. A more accurate and realistic trajectory can be predicted by considering the mass and elasticity of the cables. It is possible to extend the static workspace of the moving platform in a controlled manner by using the equivalent spring model of the cables. By calculating the maximum permissible tension in each cable, it is ensured that the tension of each cable will not exceed this maximum threshold during the trajectory. In this way, the robot has a significantly larger effective workspace and is able to perform tasks that aren't possible with traditional trajectory planning. An RRPM robot with eight cables was designed for simulating the underwater forces of a walking humanoid robot [108]. While avoiding any perturbations to the cables, the simulator robot followed the desired trajectory. A 4–3 CRPM trajectory was presented for sensing and mapping aquatic environments [88]. The robot's path planning was designed to minimize its energy consumption when manoeuvring on a repetitive path. Minimizing the vector of cable tension's second norm had this effect.

It is possible for cables to wrap around obstacles or become entangled with one another in a cluttered environment. The literature studies optimized collision-free path planning so as to address this problem. For the path planning of an IRPM in a polygonal obstacle area, a homotropy signature was used [183]. When two trajectories connect the same start and end points and can constantly deform into each other without crossing each other, they are homotopic. The shortest path between two points can be found using this approach. In path planning, piecewise linear interpolation is useful because it allows the robot to move through cluttered and complex environments without colliding with obstacles or cables [184]. Moreover, the implementation is relatively straightforward, as it does not require complex calculations or modelling of robot dynamics. The method enables the target points to be connected in a manner that prevents collisions with obstacles. A 7-6 CRPM was introduced to assist with upper limb rehabilitation, and the goal of this study is to find the appropriate CDPR for enabling patients to perform required tasks [74]. In order to achieve all the prescribed poses, it was proposed that a reconfigurable robot with moveable motor locations be used. In order to select the robot structure capable of producing minimal actuated torque along the prescribed workspace, particle swarm optimization was applied. The criteria were mathematically defined using two formulations. Normalized tension is measured in one term, while a normalized distance is measured between the end-effector's centre of mass and the cable attachment locations on the base platform. To guarantee that the robot would work properly, a set of constraints was chosen. To prevent slack cables, cable tension must be always positive. Furthermore, it cannot exceed the actuator's mechanical characteristics. Moreover, collisions with the platform or other cables were taken into account when the end-effector was in motion. Even while this approach provides better outcomes than traditional CDPRs, it still uses a large robot design unsuited for rehabilitation. A combination of a large translational workspace of CDPRs and a large rotational workspace of serial manipulators [72] was suggested as an extension of the discovered technique to enhance the workspace.

As part of Ref. [23], the high-dimensional continuation method [185] was applied to 6-6 IRPM path planning. In this method, two configurations are systematically transitioned while cable tensions remain within acceptable limits. The cables cannot become excessively slack, to prevent the robot from losing control, nor excessively tight, to prevent cable breakage even when the wrench is subject to bounded perturbations. An approach to path planning that accommodates all such constraints. Most CDPR applications involve direct human interaction, which makes safety a key consideration. A cable failure detection strategy for 4-3 CRPM upper limb rehabilitation was proposed in Ref. [65] so as to avoid any reactive motion of the end-effector resulting from the failure. Therefore, the study aims to minimize the consequences of cable failures in CDPRs and propose a methodology for planning and achieving a motion for reaching a safe position after a cable failure. By using a performance index called Wrench Exertion Capability (WEC), the force can be computed before and after a cable failure in a specific direction. To stop an end-effector, the maximum exertable braking force can be calculated using this index.

An alternative to avoiding cable slack and avoiding obstacles is to change cable configurations on moving platforms or base platforms. A modular mobile platform that can be reconfigured to perform pick-and-place operations in cluttered zones was introduced [25]. The platform configuration can be altered to prevent collisions with obstacles while reducing the period of movement. Cable connection points on the base platform may be positioned at

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any of a wide range of discrete locations, which ensures minimal cable interference. A conceptual map was created to identify the minimum sets of configurations required for accurately tracking a predefined path. This mapping process is aimed to analyse and optimise the system's performance in adhering to the specified trajectory.

The domain of spatial CDPR path planning has undergone substantial progress in overcoming the constraints of conventional approaches. However, the question of immediate adaptability has not been adequately addressed. The implementation of algorithms which can incorporate real-time sensory feedback and dynamically modify the trajectory of the robot has the potential to greatly improve the efficiency and dependability of spatial CDPRs in real scenarios.

3.4 Control of CDPRs

In comparison to rigid-link parallel robots, CDPRs have a more difficult time controlling since cables are involved. It is therefore essential that control laws are designed to maintain all tensions positively while achieving the desired control performance [186]. Further, the low stiffness of such mechanisms makes it difficult to accurately control the end-effector's position and orientation. Due to the special characteristics of cables, some frequently applied control approaches have to be modified [1]. In CDPRs, both dynamic and kinematic controllers are implemented. In kinematic control techniques, reference values corresponding to the assigned end-effector trajectory are fed to the cables using inverse kinematic transformations [187]. There is a consideration of the nonlinear dynamics of the robot when designing dynamic controllers. Table 4 provides a comprehensive list of research studies categorized based on their configurations, DOF, and arrangement with the control system, considering factors such as mass and elasticity.

3.4.1. Research on Control Assuming Non-Flexible Cables without Mass

When it comes to kinematic controllers, an IRPM that tracks spatial objects was controlled with a proportional derivative (PD) controller [89]. The position errors were measured using feedback from the IMU, encoders, and gyro sensors. A reconfigurable planar 4-3 CRPM was designed for the rehabilitation of lower extremities. An essential feature of this design was the use of a PID force servo system, which enhanced the loading accuracy and speed [54]. Based on the results of the evaluation, the force servo system was found to have a reliable capacity to operate in the standard rehabilitation frequency, thereby fulfilling the necessary criteria for effective rehabilitation. As a potential means of improving CDPR loading accuracy and speed for lower limb rehabilitation, a force servo system presents a promising prospect. In Ref. [55], the two-level controller was designed for a 3-2 planar CRPM intended for the rehabilitation of upper extremities. The lower part of the controller intricately managed the tension distribution through three cables, whereas the high-level controller calculated assistive forces. In order to improve movement accuracy and execution speed, the assistive force generates a virtual trajectory space around the intended trajectory using impedance control. In Ref. [19], PID as position controller was applied on T-Bot for a high-speed pick and place task. The forward kinematic equation is derived using the closet set-search method, and the model reference adaptive controller (MRAC) is designed to track the desired trajectory [64]. The controller was selected so that controller gain tuning could be avoided.

As far as dynamic controllers are concerned, the linear and quadratic programming techniques of feedback linearization (FL) were used to ensure positive tension in cables [186,188]. It incorporates gravity, inertial matrices, and position errors for the control input. In Ref. [189], FL control was augmented by adding a reference governor to enhance the performance of a 6-DOF non-redundant cable-suspended robot. The reference governor generates admissible reference signals under the receding horizon strategy. As a result of this approach, it was possible to predict the future state of the system using error dynamics. To compensate for the oscillatory motion of the nonlinear system, the controller reduced its phase trajectory [190]. As an indicator of the oscillatory nature of the nonlinear cable suspended 4-3 CRPM, the oscillation number was defined, and the genetic algorithm was applied to compute torque controller gain to reduce it. There is a possibility that CDPRs may interact with a disturbed environment [191]. Furthermore, there exist various uncertain physical properties of CDPRs that could affect their performance, such as their mass, the end-effector's moment of inertia, and the diameter of the cable pulley. To achieve more accurate trajectory tracking, adaptable and robust controllers should be employed. For a planar 4-3 CRPM, a robust PID controller was developed to handle structured and arbitrary deviations in the parameters of the robot, such as the end-effector's mass and inertia [192]. Positive tension was maintained in the cables by using the vector of internal tension that was derived from the null space of the transposed Jacobian matrix of the robot. A comparative analysis of a vision-based system provided feedback for the implementation of three different control algorithms: pole

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placement, sliding mode, and adaptive sliding mode was presented for planar 4-3 CRPM [193]. The sliding mode control exhibits a unique phenomenon called chattering and is difficult to get rid of. However, the analysis indicated that adaptive sliding-mode controllers were best at handling uncertainties with regard to chattering. 's_k' is defined as the sliding surface which corresponds to $s_k = k - k_{ref}$. The elements 'k' and 'k_{ref}' corresponds to the position and orientation parameters of the end-effector. A 6–6 IRPM was controlled with an adaptive control algorithm and FL control [194]. A comparison of the results demonstrated that the adaptive controller is effective at compensating for uncertainties in terms of mass and moment of inertia. A spatial 6-6 IRPM carried by a helicopter was controlled by an adaptive controller somewhat similar to inverse dynamic approach [45]. The robot carried out both translational and rotational motions in order to compensate for the disturbances created by helicopter motion. As a result of this method, the cable robot is stabilized irrespective of the helicopter's motion with the consideration of uncertainties such as variation in mass and moments of inertia. A study introduced a potential energy shaping for method for underactuated parallel robots using the interconnection and damping assignment passivity-based control (IDA-PBC) method. Addressing the challenge of fewer actuators than DOF, a set of partial differential equations (PDEs) is solved to assign a new Hamiltonian to the closed-loop system, a key aspect of the IDA-PBC approach. The proposed approach stabilizes a natural equilibrium point of the system, applicable to 3-6 IRPM. The results are compared with sliding mode and joint space control. Homogeneous solutions of the potential energy PDE are joint space variables, offering a clear physical interpretation when designing the desired potential energy. In a later study, the proposed approach is generalised for potential energy shaping for all CDPRs with any degree of underactuation [195].

An interactive control system can function with either an impedance or an admittance controller. The impedance control generates an output force based on displacement input in the dynamic model. Conversely, admittance control generates displacements based on measured forces. An adaptive control incorporating admittance control was employed to avoid possible collision between cables during translational motion in a spatial CRPM [47]. When cables were close to colliding, opposing forces were generated. Karush–Kuhn–Tucker conditions were used to calculate the minimum distance between two cables. An 8–6 suspended RRPM was controlled visually by 2½D visual servo control [196]. A vision algorithm provided feedback to the trajectory tracker in order to make the system more robust. The 4–2 RRPM was controlled using a continuous-time generalized predictive control [197]. A quadratic cost function was minimized over a finite horizon in continuous time. Considering cable tension constraints, the cost function is defined as the difference of predicted positions and the desired trajectory. PD and sliding mode controllers were used to control an 8–6 suspended RRPM [198]. As far as accuracy and repeatability are concerned, the controller provided an adequate response given the uncertainties associated with the end effector's mass.

In Ref. [71], a neural network controller featuring an adaptive multilayer neural network was introduced for a 4-2 IRPM, designed to mitigate perturbations. This controller incorporated a term into the central control system, accounting for the dynamics of actuators and gearboxes, providing a cable tension command within a defined range. Extending this research theme, another noteworthy study detailed in Ref. [56], involved the implementation of a neural network controller designed for a 4–3 planar rehabilitation robot. This controller allowed for the pursuit of a desired trajectory by permitting an adjustable tracking error. This feature permitted human subjects to move their limbs freely within this error margin, with the added benefit of an adjustable assistance level. The underlying algorithm not only facilitated trajectory tracking but also contributed to enhanced rehabilitation efficiency. Another study [199] proposed an adaptive control law for upper limb rehabilitation of 3-6 IRPMs. This approach incorporated a fuzzy tuner to adjust control parameters based on position errors. It considered external uncertainties associated with the environment during robot-assisted rehabilitation, as well as human dynamic properties over time. Additionally, an adaptive fuzzy tracking control strategy was outlined for trajectory tracking control in the presence of external disturbances for a 4-3 CRPM [39]. A sort of hybrid control strategy PID integrated with feedforward Neural Network was introduced [73]. Using the FNN, a wrench in the task space can be mapped to a set of cable tensions in the joint space. As for RRPMs, achieving effective cable tension distribution in real-time, particularly with an embedded algorithm, necessitates a suitable control scheme. Various redundancy resolution approaches are employed to ensure the cables maintain a positive tension. In Ref. [200], a non-iterative real-time compatible method for distributing tension was implemented as part of a dual-space feedforward scheme. The method allows tension distributions to be made in real-time without requiring iterative processes. The issue of redundancy resolution in a 4–2 RRPM was addressed using iterative method in Ref.

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[201]. This approach formulates a convex optimization problem, incorporating constraints such as the structural characteristics of the manipulator and cable dynamics. The analysis of the optimization problem involves the utilization of the Karush–Kuhn–Tucker theorem, and for redundancy resolution, a straightforward iterative search algorithm was proposed. A cost function defined on the p-norms of cable tension was used to calculate tension distributions for an 8-6 RRPM [128]. This strategy ensures continuous and differentiable tension distributions along trajectories. A similar optimization approach was adopted for a 3-2 RRPM planar CDPR [202], demonstrating the versatility of this method across different RRPM configurations. In Ref. [203], a unique approach to optimal tension distribution in various CDPR configurations was introduced, utilizing a flexible cost function that allows for simple tuning. This cost function is comprised of fixed logarithmic barrier and p-norm functions, carefully chosen to prevent unwanted discontinuous accelerations of actuators and ensure continuous differentiability of actuator forces. This method addresses the challenge of achieving optimal tension distribution while maintaining smooth system operation. By taking cable pretension into consideration, a passive plant for planar RRPMs was set up in Ref. [204], leading to the development of a strictly positive controller. In order to ensure precisely positive tension in the cables, this controller uses a method that avoids actuator saturation and depends on an approach to a problem with linear programming.

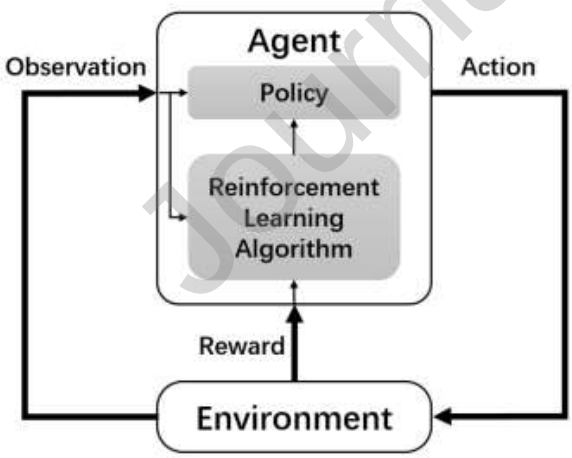

Figure 12. The schematic diagram of reinforcement learning

A model-based predictive control technique was employed for RRPMs in Ref. [205], with a particular emphasis on cable tension boundary constraints. This approach integrates cable tension distribution seamlessly into the primary model-based control architecture. By explicitly addressing tension constraints, the control strategy ensures optimal cable tensioning within the system, enhancing RRPM performance. In Ref. [206], a unique approach for achieving positive tension distribution in CDPRs, without resorting to redundancy resolution approaches, was introduced. The controller is integrated with a positive saturation-type function to map control efforts into a positive range, effectively addressing tension requirements. To counteract the effects of saturation, a nonlinear disturbance observer is integrated into the system. The stability of this innovative control scheme is rigorously verified through Lyapunov's theorem, ensuring robust and reliable performance. For the position tracking of CDPRs, a nonlinear model predictive control (NMPC) strategy was proposed later [40]. A cable tension and its limit are explicitly handled by the NMPC formulation. A significant component of NMPC feedback control strategies is cable tension distribution, notably allowing the CDPR to operate effectively across wrench-feasible workspace boundaries.

A subfield of machine learning, known as reinforcement learning (RL), has been extensively used in the field of robotics [207–209]. They have been utilized with robot navigation, gaming and taking real-time decisions [210]. It is concerned with a fundamental issue of how intelligent systems learn to make a good sequence of decisions as continuous optimal solutions [211]. For learning, system doesn't know in advance how its decisions are going to affect the real world or what decisions might necessarily be associated with good outcomes and instead it has to acquire that information through experience [212]. ADP stands for approximate dynamic programming, which is the principle behind RL. An RL agent observes the environment at every instant, selects an action according to the policy, and receives an immediate reward based on a reward function. An agent interacts with a complex and uncertain situation in order to maximize their cumulative reward. By interacting with the environment through experience, RL allows robots to determine an optimal decision-making strategy. Figure 12 illustrates the schematic of an RL agent observing and interacting with an environment in form of an action against a defined policy in order to collect maximum reward [66].

Two methods of RL have been suggested in the literature to determine the optimal policy. An optimal value function and policy are determined using Q-learning and deep Q-networks (DQNs). By contrast, policy-based methods, such as deep deterministic policy gradient (DDPG), asynchronous advantage actor−critic (A3C), and twin delayed DDPG (TD3), directly determine optimal policies. An RL algorithm combining Q-learning with deep neural

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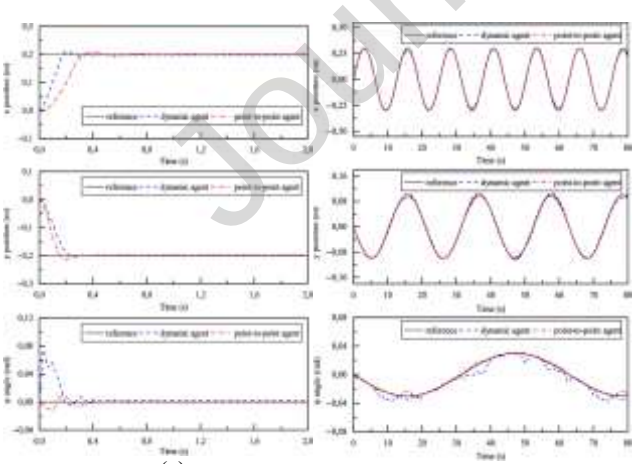
networks also known as DQN is proposed for implementation on a 4-3 CRPM for rehabilitation purposes [66]. Increasing training time decreases both the total tension of the four cables and the deflections in trajectory. The Q-learning algorithm is inadequate for applications where the state and action space is extensive in terms of limited memory and large computational costs. In addition to high-dimensional scenarios, the DQN method developed using ANNs instead of tables can address continuous-state space scenarios. Despite the fact that DQN is capable of solving problems with high-dimensional observation spaces, it can only handle discrete and low-dimensional action spaces. Physical processes, on the other hand, often have a large and continuous action space. For high-dimensional problems with such continuous action structures, methods such as DDPG, A3C, and TD3 are used. In Ref. [120], RL was introduced for point-to-point movement and reference dynamic tracking of a planar 4-3 CRPM with one rotational motion. To train the RL agents in point-to-point and dynamic reference tracking tasks, DDPG is used. A total of four artificial neural networks are used in the DDPG method: the Critic-Network, the Actor-Network, the Target Critic-Network, and the Target Actor-Network. The response of the proposed strategy is indicated in figure 13a showcasing the response to step task while figure 13b showcases the response to dynamic task [120]. In order to maximize long-term reward, the actor observes the current situation and determines a course of action. Critics, on the other hand, take observations and actions as input and estimate the long-term reward value. A novel reward function was designed to maximise the expected reward which is given by:



(a) (b)

Figure 13 Response: (a) Response 1: Control response of the dynamic and point-to-point agent for the point-to-point (step) task, (b) Response 2: Control response of the dynamic and point-to-point agent for the dynamic reference tracking task

$$x_b = F_x\left(\frac{l}{EA} + \frac{\sinh^{-1}\left(\frac{F_z}{F_x}\right) - \sinh^{-1}\left(\frac{F_z - \mu g l}{F_x}\right)}{\mu g}\right). \tag{8}$$

Specifically trained tasks are used to evaluate the performance of three two agents. The same agent is also implemented on dynamic reference tracking tasks and vice versa. An analysis of the performance of RL agents is compared with the performance of a classical PD controller. It was indicated that through its inherent optimization of cable tensions, the method eliminates the need for tension distribution algorithms to control a system's dynamics.

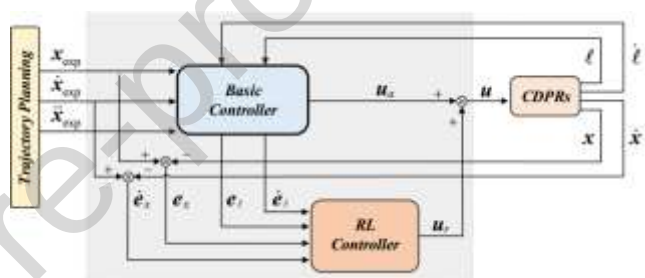


Figure 14. Control diagram of reinforcement learning control algorithm

In Ref. [75], it was examined how the integration of DRL with non-learning-based methods impacts learning speed and the robustness of DRL to model uncertainties. A CDPR is controlled by a hybrid DRL strategy. Using a robot involves decomposing the task of manipulating it into multiple subtasks. A subtask can be accomplished using inverse dynamics equations or PID algorithms. DDPG and the inverse dynamics equation of the CDPR are combined to form the hybrid DRL strategy. In Ref. [213], a DRL control for a 3-3 CRPM was introduced. The proposed learning control framework for CDPRs is based on a Lyapunov-based off- policy soft actor-critic algorithm [214]. Figure 14 illustrates a visual representation of the scheme that incorporates RL alongside the fundamental controller [213]. This integration facilitates a dynamic and responsive control strategy, enhancing the system's performance and adaptability to varying conditions.

Figure 15 presents a detailed comparison of tracking performance across various trajectories, encompassing both oblique and flat circle trajectories as well as oblique and flat eight-type trajectories [213]. These trajectories are analysed under three distinct cases: expected trajectory (EXP), basic controller with accurate model parameters (BS), basic

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controller with uncertain model parameters (BU), and RL-based controller with uncertain model parameters (RLU). Figure 15 (a)–(d) showcase the comparative tracking performance of these controllers, providing insights into their efficacy under different conditions . This comprehensive analysis enables a thorough examination of the controllers' capabilities across a range of trajectories and uncertainty levels, facilitating a deeper understanding of their performance characteristics.

The generalisation of the findings in these investigations is limited due to the dependence on assumptions of non-flexible, massless cables. Future research should consider the impact of cable flexibility and mass to develop more robust and adaptable control strategies. Moreover, experimental validation in real-world settings is crucial for advancing the practical applicability of these control methods.

A closed-form algorithm for force distribution in parallel wire robots was developed, addressing one of the key challenges in controlling redundantly actuated cable-driven systems. The approach offers a fast and practical way to compute wire forces that remain feasible, continuous, and differentiable along trajectories, making it especially well-suited for real-time robotic applications [215].

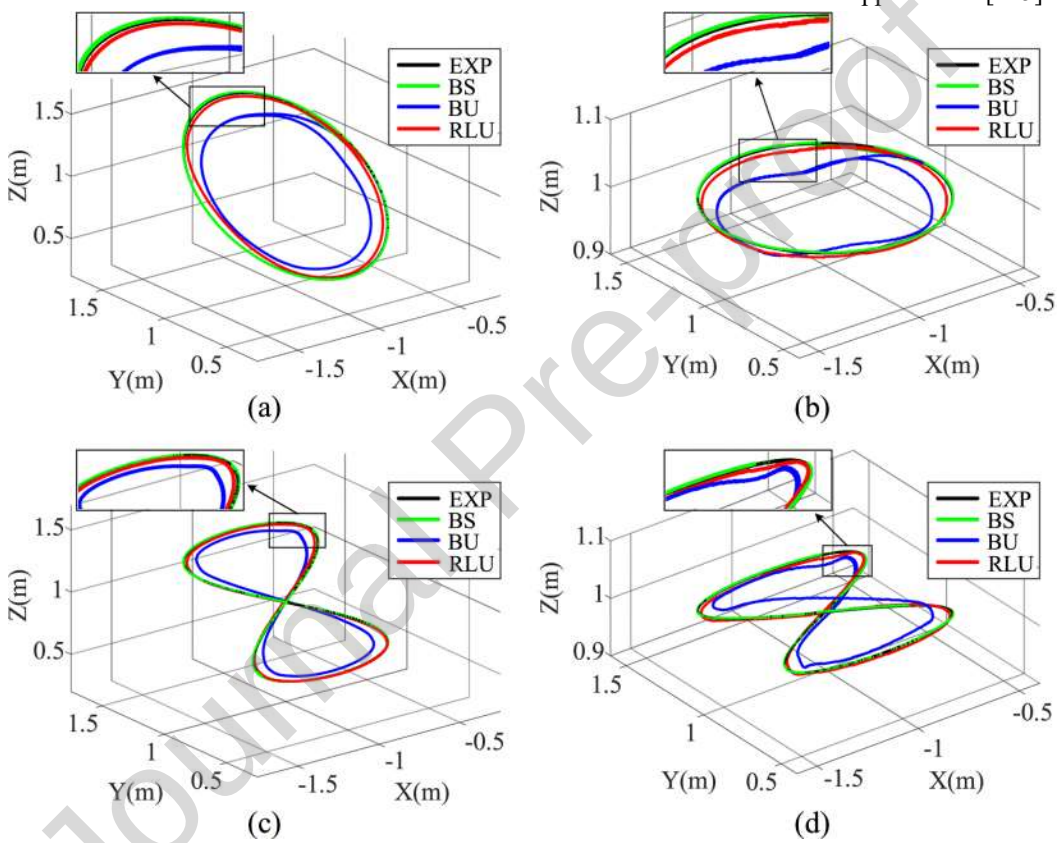

Figure 15 Tracking Comparison diagrams in (a) Oblique Circle Trajectory, (b) flat circle trajectory, (c)Oblique eight-type trajectory and (d) flat eight-type trajectory under three cases [EXP-expected trajectory, BS- Basic Controller when model parameters are accurate, BU-Basic controller when model parameters are uncertain, RLU- RL based controller when model parameters are uncertain]

Table 4 Control of CDPRs

Research on Control Assuming Non-Flexible Cables without Mass			
Name/ Ref./ Application	Control system	Configuration and DOF	Arrangement (Planar/ Spatial)
T-Bot [19]	PID based position control	6-3 RRPM	Planar
Liu et al. [39]	adaptive fuzzy tracking control strategy	suspended 4-3 CRPM	Planar

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HRPCable, Santos et al. [40]	Position tracking using nonlinear MPC	8-6 RRPM	Spatial
Manipulator suspended with a helicopter [45]	Adaptive controller to stabilize the manipulator with disturbance in movement of the helicopter	Cable suspended 6-6 IRPM	Spatial
Meziane et al. [47]	Adaptive control using admittance controller to eliminate cable interference in translational motion	7-6 CRPM	Spatial
Lower limb Rehabilitation [54]	PID based servo system	Reconfigurable 4-3CRPM	Planar
Upper limb Rehabilitation [55]	Two level control. a. Low-level used for tension distribution. b. Impedance control used for assistive force.	3-2 CRPM	Planar
Rehabilitation robot [56]	Controller with adaptive tracking error adjustment using neural networks (adaptive NNC)	4-3 CRPM	Planar
Kim et al. [64]	Model reference adaptive control	3-2 CRPM	Planar
Xie et al. [66]	DQN based reinforcement learning	4-3 CRPM	Planar
Jabbari Asl et al. [71]	Neural network controller to compensate perturbations	4-2 IRPM	Planar
Xiong et al. [73]	PID integrated with feedforward neural network	4-3 CRPM	Planar rotational
Xiong et al. [75]	Hybrid DDPG integrated with inverse dynamics of CDPR	4-3 CPRM (Similar platform as)	Spatial (Rotational only)
Tracking of Spatial Objects [106]	PD	Cable suspended 4-6 IRPM	spatial
Sancak et al. [120]	DDPG based RL	4-3 CRPM	PLanar
IPAnema [128]	Continuous tension distribution using p-norms	8-6 RRPM	Spatial
Oh et al. [186]	Feedback linearization (FL) to ensure cables are positively tensioned	Cable suspended RRPM	Planar
Oh et al. [188]	Feedback linearization (FL) to ensure cables are positively tensioned	RRPM	Planar
Oh et al. [189]	Feedback linearization (FL) augmented with reference governor enabling to predict future error dynamics	Cable suspended 6-6 CPRM	Spatial
Zarei et al. [190]	Phase trajectory reduction to reduce oscillations using genetic algorithm compute torque controller (PD) gains	Cable suspended 4-3 CRPM	Planar
Khosravi et al. [192]	Robust PID developed to handle structured and unstructured uncertainties	4-3 CRPM	Planar
Bayani et al. [193]	pole placement, sliding mode, and adaptive sliding	4-3 CRPM	Planar

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	mode		
Zarebidoki et al. [194]	Adaptive and FL	Cable suspended 6-6 IRPM	Spatial
Zake et al. [196]	2½D visual servo control with Lyapunov stability analysis	7-6 CRPM	Spatial
Inel et al. [197]	Continuous time generalized predictive control	4-2 RRPM	Planar
Picard et al. [198]	PD and sliding mode	Suspended 8-6 RRPM	Spatial
Upper Limb Rehabilitation [199]	Fuzzy Inference used to adjust controller parameters	3-6 IRPM	Spatial
Lamaury et al. [200]	Dual space feedforward	Suspended 8-6RRPM	Spatial
Taghirad [201]	Tractable and Iterative search algorithm is used to resolve redundancy (cable tension Distribution)	4-2 RRPM	Planar
Gosselin et al. [202]	p-norm (p=4) to optimize the tension distribution	3-2 RRPM	Planar
Ueland et al. [203]	Optimization for tension distribution (flexible cost function)	4-3 CPRM	Planar
Caverly et al. [204]	Strictly positive controller	4-2 RRPM	Planar
HRPCable, Santos et al. [205]	Model predictive controller (MPC) handling cable tension limits	8-2 RRPM (Designed to work in spatial configuration)	Planar (arranged for spatial)
Ameri et al. [206]	Non-linear observer based sliding mode controller	8-6 RRPM	Spatial
Lu et al. [213]	Lyapunov based soft actor-critic RL	3-3 CRPM	Planar
IPAnema [215]	Continuous tension distribution using p-norms ($2 \leq p \leq \infty$)	8-6 RRPM	Spatial
	Research on control assuming Hefty cables		
Duy et al. [38]	PD torque controller	Suspended 8-3 RRPM	Planar
Mobile wheeled CDPR [46]	Feedback linearization	Suspended 6-6 CRPM	Spatial
FAST telescope [52]	PI integrated fuzzy control	Suspended 6-6 CRPM	Spatial
FAST telescope [216]	Hybrid force/ Position PID controller	Suspended 7-6 CRPM	Spatial
Square kilometre array [217]	Fuzzy controller	Suspended 6-6 CPRM	Spatial
Caverly et al. [218]	Comparative analysis of PD and filtered error control (derivative control law)	2-1 CRPM	Planar

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Research on control assuming flexible cables			
Zhang et al. [58]	PI control	3-3 IRPM	Planar
Jamshidifar et al. [90]	$H\infty$	12-3 IRPM	Planar T-R
Jamshidifar et al. [91]	PD controller	8-6 RRPM	Spatial
CoMIRO [93]	Pole placement for gain scheduling	Suspended 8-6 RRPM	Spatial
Korayem et al. [139]	Feedback Linearization (FL) with LQR and LQG for estimating process noise	6-6 IRPM	Spatial
Korayem et al. [140]	FL with PD controller, stability analysis using Lyapunov criteria	6-6 IRPM	Spatial
Raman et al. [172]	TD3	4-3 CRPM	Planar
KNTU galaxy [192]	PD, Lyapunov theorem for stability	8-6 IRPM	Spatial
Yang et al. [199]	Fuzzy PID	4-3 CRPM	Planar T-R
Babaghasabha et al. [219]	PD, Lyapunov theorem for stability	4-3 CRPM	Planar
FALCON [220]	PD	7-6 CRPM	Spatial
Aghaseyedabdollah et al. [221]	Interval type-2 fuzzy sliding mode	4-3 CRPM	Planar with extra DOF for base rotation
Wang et al. [222]	HER (hindsight experience replay) based soft actor-critic RL	reconfigurable 3-3 IRPM	Planar with extra DOF for base rotation

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3.4.2. Research on Control Assuming Hefty Cables

The control strategies discussed in this subsection are designed to address the challenges posed by hefty cables such as sagging of cables during robot dynamics, with the primary goal of compensating for their effects to achieve precise positioning. One of the effects when considering cables with mass is sagging of cables. In Ref. [216], an innovative hybrid position/force PID controller was introduced, specifically tailored to counteract the pseudo-drag problem associated with flexible cables in the FAST system. The designed control system incorporates feedback from both cable force and position, enabling a comprehensive approach to cable dynamics. The tracking error achieved through the implementation of this controller is notably low, measuring approximately 2 mm. This level of precision satisfies the stringent accuracy requirements for the FAST system. In the study presented in Ref. [46], a mobile wheeled CDPR was introduced. The dynamic equation, incorporating cable mass, was derived using the Gibbs–Appell method. This equation played a crucial role in the feedback linearization (FL), effectively eliminating system nonlinearities. The application of fuzzy logic in controlling CDPRs with substantial cables was explored, involving the tuning of parametric control gains. A sophisticated fuzzy Model Reference Learning Controller was proposed in Ref. [217]. This controller was designed for a large workspace robot and comprised three main components: a direct fuzzy controller, a reference fuzzy inverse model, and a fuzzy learning mechanism. The direct fuzzy controller was fine-tuned to ensure that the closed-loop system mimicked a prespecified reference model, facilitated by the fuzzy learning mechanism. In the

investigation documented in Ref. [52], the FAST underwent an inverse dynamics analysis through the Lagrangian method. Simulating real-world conditions, random wind forces acting on the end-effector were replicated. Effectively addressing wind-induced vibrations, a fuzzy based PD controller was implemented. Ref. [218] introduced a lumped-mass dynamic model of a planar 2-1 CRPM with a passivity-based controller. The dynamic model accounts for the varying cable stiffness and mass as the cable winds around a winch, along with changes in winch inertia during this process. The assumption of the payload's significant amount of mass than the cables and winches allow for decoupling of rigid and elastic dynamics. This approximation facilitates the definition of modified input torque and output rate, establishing passive input-output mappings. Furthermore, Ref. [38] implemented a PD torque controller on a suspended planar 8-3 RRPM, showcasing the versatility of control strategies in different cable-driven parallel robot configurations.

Though the current research provides valuable insights into controlling CDPRs with hefty cables, there is a need for more adaptive control strategies that can respond to real-time changes in cable dynamics. Future studies should explore the integration of advanced sensing technologies and adaptive algorithms to enhance the robustness and accuracy of control in dynamic environments.

3.4.3. Research on Control Assuming Flexible Cables

Various strategies are implemented to mitigate the impact of elasticity and vibration on CDPRs. One such strategy involves segregating the quick and slow motions of the system. In Ref. [138], the dynamic equations of a spatial independently rotating platform mechanism (IRPM) were reformulated using the singular perturbation approach, transforming them into the standard form. To ensure the stability and robustness, a corrective term $K_d = (\dot{L}_1 - \dot{L}_2)$ was introduced to the rigid part of the controller. Here, L_1 and L_2 represent the vectors of cables under tension, and K_d is a diagonal matrix. The Tikhonov theorem was employed to distinguish between slow and quick variables, facilitating stability analysis. In Ref. [219], a comprehensive and robust adaptive approach was introduced for a 4–3 planar CRPM. In order to counterbalance longitudinal vibrations, a similar control corrective term was incorporated. Expanding on this, Ref. [90] addressed the undesired vibrations of the end-effector by segregating them from the desired equation of motion. This was achieved by omitting second and higher-order motion errors, resulting in a linear parametric variable dynamic system. A robust vibration compensator, developed using the $H\infty$ approach, was employed to govern the system. In Refs. [199,220], PD and PID controllers, along with fuzzy based adaptive PID controller, were employed for a 7–6 spatial CRPM and a 4–3 planar CRPM, respectively. The fuzzy method played a crucial role in fine-tuning the gains of the PID controller. The fuzzy method played a pivotal role in the optimization of the PID controller gains. Moreover, an online dynamic minimum torque estimation mechanism was incorporated to guarantee positive tensions in the cables. Notably, the system responses indicated that the fuzzy PID controller exhibited superior robustness in comparison to the PID controller, particularly in the face of disturbances stemming from the elastic cables. A supervisory interval type-2 fuzzy adaptive sliding mode control scheme for 4-3 CRPM, combining intelligent methods with conventional sliding mode control to optimize control parameters was introduced [221]. Addressing the structural constraints of cable robots, including purely tensile forces and inherent elasticity, the proposed approach ensures accurate tracking performance. Utilizing the internal force concept and the singular perturbation theorem, the controller handles the effects of elastic cables, ensuring stability through the second Lyapunov method. An interval type-2 fuzzy logic controller adjusts control gain to reduce chattering, while a supervisory interval type-2 fuzzy logic control system regulates gains within the sliding surface. An interval type-2 fuzzy sliding surface is optimized using the grasshopper optimization algorithm (GOA). Results from simulations demonstrate desired tracking performance despite uncertain parameters and structural constraints.

In the exploration of controlling the effects of elasticity and vibration on CDPRs, various strategies have been employed. In Ref. [140], inertial measurement units (IMUs) and encoders were integrated to discern fast and slow dynamic movements of the end-effector. The feedback from these sensors was then utilized in a feedback linearization (FL) controller, where the precise position and orientation of the end-effector. were estimated. The Kalman filter was employed as the foundation of a data fusion method, ensuring noise-free determination of the end effector pose. Ref. [139] considered end-effector vibration as process noise. In order to strike a balance between control input and tracking error, FL gains were determined using the linear quadratic Gaussian method. A Lyapunov analysis illustrated that a system with damping less than a specified minimum value could maintain stability with this approach. For non-model-based vibration control of CDPRs, [91] introduced a mechanical reaction-based stabilizer. This stabilizer, consisting of three actuators attached to a pendulum in a perpendicular arrangement mounted on the end-effector, required only the actuators' directly measurable position and

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velocity to form closed-loop control feedback signals. In Ref. [93], a modal analysis approach was employed to formulate a linear decoupled model for an 8–6 suspended RRPM featuring elastic cables. The resultant model, when projected into modal space, produced six decoupled second-order differential equations corresponding to the robot's six DOF. This configuration allowed for control using the conventional SISO method. An obstacle avoidance problem for a reconfigurable 3-4 IRPM is solved using DRL. Using hindsight experience replay (HER), the improved soft actor-critic algorithm is applied to train the robot to avoid obstacles in the reconfigurable CDPR environment. In a study, three distinct approaches for implementing model-free DRL in dynamic trajectory tracking were explored [172]. The first approach involves applying DRL to a conventional non-redundant CDPR operating within a fixed workspace. The scenario allows for the investigation of DRL's efficacy in optimizing trajectory tracking performance without redundancy. The second approach delves into an end-to-end approach, incorporating redundancy resolution in a reconfigurable CDPR [222]. This setting explores the potential of DRL to seamlessly handle the complexities of reconfigurable robots with redundancies, offering a comprehensive solution for dynamic trajectory tracking. The third approach adopts a decoupled approach, focusing on resolving kinematic and actuation redundancies independently. This approach allows for a nuanced exploration of DRL's capabilities in addressing specific types of redundancies within the CDPR, potentially enhancing the overall efficiency of trajectory tracking.

The current research has provided a strong foundation for controlling flexible cables in CDPRs, but there is a need for further studies to evaluate these approaches in practical situations. Additionally, research should focus on developing generalized control approaches which can be applied across various CDPR configurations, ensuring broader applicability.

4 Conclusions

CDPRs have been extensively examined in this comprehensive review, both in terms of their planar and spatial configurations and their applications. The survey highlighted the significance of new and emerging research directions, particularly in the context of CDPRs.

1. With particular emphasis on the consequential effects introduced by considering the cable's mass and elasticity, with implications extending to kinematics, dynamics, and control strategies. However, this survey identifies several noteworthy research gaps and areas for future exploration within CDPRs. Firstly, while existing literature predominantly considers linear behaviour for cables, there is a notable lack of investigation into the mechanics, as well as the forces and torques involved during motion, with due consideration to the nonlinear elastic characteristics of flexible cables. The formulation of a comprehensive framework incorporating mass, nonlinear flexible behaviour, and environmental factors remains an avenue for future research. Additionally, few studies have explored the impact of cable mass and elasticity on workspace and trajectory planning, necessitating further exploration and analysis.
2. Addressing the need for configuration optimization, the review explored methodologies such as motor and cable placement adjustments and the incorporation of additional elements. These strategies aim to enhance the operative area and facilitate unhindered trajectory path.
3. Control systems that are robust and adaptive were also discussed, offering solutions to counteract uncertainties in robot parameters and external disturbances. A key aspect of the review relates to the emerging stage of merging CDPRs with traditional serial and parallel robots to create hybrid manipulators. While CDPRs can enhance the DOF in such systems, comprehensive investigations into kinematics, dynamics, workspace, trajectory planning, and control in these configurations are essential for their successful implementation [72].
4. The literature has also paid limited attention to CDPR applications in hydrodynamic fluid and low gravity environments. Given the inherent advantages of a large operational area and lightweight structure, CDPRs present an appealing and viable option for such applications. However, further research is required to consider implications of fluid forces in dynamics analysis and control implementation, such as upthrust and fluid resistance, i.e., friction, on the end-effector and cables. Developing algorithms specifically tailored to these applications is paramount to handling the nonlinear dynamics associated with hydrodynamic forces. Future research directions include leveraging the method for robust calibration and dynamic parameter identification of CDPRs, modelling cable deformation and sagging to address vibrational effects, and exploring alternative techniques for nonlinear system approximation to better capture system behaviour over large oscillations.

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5. The reviewed literature extensively investigates into learning-based methods within the context of CDPRs, with a primary focus on their potential applications in the kinematics, dynamics, and control aspects of CDPR systems. ML methods are explored, revealing their efficacy in optimizing CDPR models, resulting in an impressive accuracy improvement of up to 66% and a substantial reduction in error variance [129]. Furthermore, RL is investigated emphasizing its applications in enhancing the overall performance and adaptability of CDPR systems when incorporated into the control framework. By incorporating reinforcement learning into the control system, the review suggests a synergistic relationship that can significantly improve the overall performance and adaptability of CDPR applications. One key advantage highlighted is the ability of reinforcement learning to facilitate the determination of continuous optimal control solutions. This implies that the control strategy, guided by RL algorithms, can dynamically adjust, and optimize its solutions to achieve desired system behaviour without implementing a force/torque distribution algorithm separately. This adaptability is particularly valuable in scenarios where precise and continuous control is essential. This approach aligns with the trend in robotics towards intelligent, adaptive systems that can autonomously improve their performance through learning from interactions with their environment.
6. Though many reviews provide broad overviews of CDPRs, this study explores deeply into specific areas, offering a comprehensive understanding of advancements in suppressing oscillations in trajectory generation for underactuated CDPRs and learning based intelligent control. By focusing on these intricate aspects, the review provides valuable insights into the latest research trends and developments, making it a notable contribution to the scientific community. In conclusion, the identified research gaps and future directions outlined in this review will pave the way for continued advancements in the field of Cable-Driven Parallel Robots.

Authors' Contributions

M. Kamran Joyo wrote the manuscript; Han Yuan supervised the project and assisted with writing and review; Jun Wu assisted with writing and supervised the project; Wenfu Xu assisted in critical discussion for manuscript preparation; Honghao Yue assisted in critical discussion for manuscript preparation. All authors read and approved the final manuscript.

Competing Interests

The authors declare no competing financial interests.

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